

Diagnostic Interpretation of Effective Latent Dimensionality in Autoencoders and Variational Autoencoders

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Abstract

This paper proposes a multi-indicator diagnostic framework that integrates active-unit (AU) dynamics, intrinsic-dimension (ID) estimation (TwoNN/MLE), reconstruction-error (MSE) elbows, and PCA baselines to diagnostically interpret the effective latent dimensionality actually used in the representations learned by autoencoders (AEs) and variational autoencoders (VAEs). The goal of the framework is not to determine the effective latent dimensionality as a single unique number. Rather, it explicitly separates the diagnostic roles of the indicators: AU values serve as direct observations of the number of latent coordinates a VAE actually uses; TwoNN/MLE ID estimates serve as numerically stable auxiliary reference estimators; and the AU slowdown point m^{knee} is treated as an exploratory change point subject to grid and seed dependence. Effective-dimension usage is then diagnosed from the consistency of the multiple indicators. Theoretically, under a rate–distortion surrogate of the linear-Gaussian β -VAE, we derive the condition $\lambda_j > \beta\tau^2$ for the j -th eigendirection of the data covariance to be activated, explaining the mechanism by which KL regularization suppresses effective latent dimensionality from a water-filling perspective. This analysis is positioned not as a rigorous theorem for nonlinear VAEs but as a linear surrogate model for interpreting the experimental results. Experimentally, we show that in fully connected VAEs trained on MNIST, AU values and TwoNN IDs are highly reproducible across random seeds, whereas m^{knee} fluctuates strongly on sparse m -grids. By densely searching the neighborhood of the estimated ID and evaluating with independently initialized multi-seed runs, m^{knee} stabilizes at 10.3 ± 0.9 , consistent with the TwoNN ID near 10. Furthermore, through Fashion-MNIST, Conv-VAE, and CIFAR-10/SVHN experiments, we quantify the conditions under which the proposed diagnostic framework functions effectively and the boundary conditions under which AU/ m^{knee} diagnostics become inconclusive.

Keywords: variational autoencoder; effective latent dimensionality; active units; intrinsic dimension; TwoNN; diagnostic framework; representation learning; knowledge extraction; posterior collapse

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1. Introduction

1.1. Background: The Problem of Interpreting Effective Latent Dimensionality

In representation learning, autoencoders (AEs) and variational autoencoders (VAEs) [12] are core unsupervised models that learn low-dimensional latent representations from high-dimensional data [7]. The most fundamental design parameter governing the “quality” of these latent representations is the bottleneck dimension m : if m is too small, information

is lost, whereas if it is too large, redundant dimensions arise (and, in VAEs, posterior collapse [21] worsens).

However, a methodology for quantitatively reading “how many dimensions a learned latent representation actually uses”—that is, the **effective latent dimensionality**—has not been sufficiently established in deep representation learning research. Existing studies [17,19] have measured the intrinsic dimension of individual layers of deep networks, but studies that systematically analyze the effective dimensionality at the bottleneck of AEs/VAEs together with its training dynamics remain limited.

The central question addressed in this paper is:

How do AEs/VAEs “learn” the local intrinsic dimensionality and the topological embedding dimensionality of the input data as the effective dimensionality of the latent space? And is there a way to quantitatively read this effective dimensionality from the outside?

1.2. Positioning and Contributions of This Study

To address this question, we propose a **multi-indicator diagnostic framework** built on two pillars: **intrinsic-dimension (ID) estimation** and **active-unit (AU) dynamics**. The novelty of the framework lies not in “uniquely determining” the effective dimensionality with a single indicator, but in combining the indicators while making the robustness and exploratory character of each explicit, and **interpreting them diagnostically**. We do not claim that “ m^{knee} uniquely identifies an appropriate bottleneck dimension”— m^{knee} is a coarse exploratory design guide, and the primary readout of effective dimensionality is carried by the robust indicator (AU values) and the numerically stable auxiliary reference estimator (TwoNN ID).

Roles and robustness of each indicator:

- **TwoNN/MLE ID:** numerically stable auxiliary reference estimators that infer the effective degrees of freedom from local distances (across-seed $\sigma \leq 0.12$; subject to isometry bias; Section 5.3.5).
- **AU values:** a robust indicator that directly observes whether the KL-regularized VAE actually uses each dimension (across-seed $\sigma \leq 0.5$; independent of the distance preservation of the reference AE; dependent on β , δ , and model capacity).
- **MSE elbow and PCA:** auxiliary readouts of effective dimensionality from the reconstruction-error side.
- **AU slowdown point m^{knee} :** an exploratory indicator derived from the shape of the AU curve above—it coarsely captures the turning point at which KL regularization begins compressing dimensions, but its value depends on the m grid and the random seed (Section 5.3.5).

The contributions of this paper are threefold:

1. **Proposal of a multi-indicator diagnostic framework:** The main contribution of this study is not the proposal of m^{knee} itself, but the organization of AU values, TwoNN/MLE ID, the MSE elbow, and PCA as diagnostic indicators with distinct roles, providing a framework for interpreting the effective latent dimensionality of AEs/VAEs from multiple perspectives. We clearly distinguish AU values as a robust indicator that directly observes latent-dimension usage, TwoNN ID as a numerically stable auxiliary reference estimator, and m^{knee} as an exploratory change point obtained from the shape of the AU curve (Sections 3 and 6). Three-seed verification on MNIST confirms the high reproducibility of AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12), and we present $m^{\text{knee}} = 10.3 \pm 0.9$ under dense-grid, independent-initialization, multi-seed conditions (Exp-Mnist-Dense-300) as the main stabilization result (Section 5.3.7).
2. **Derivation of the AU activation condition based on a linear-Gaussian rate-distortion surrogate:** Under the linear-Gaussian β -VAE rate-distortion surrogate,

we derive the condition $\lambda_j > \beta\tau^2$ for the j -th latent dimension to be active, using the water-filling theorem and the KKT conditions (Theorem 1, Section 4), and explain the mechanism by which KL regularization suppresses effective dimensionality from the perspective of the water-filling principle. This result is not a rigorous theorem for actual non-linear VAEs; it is an analytical result based on the linear-Gaussian rate-distortion surrogate that provides a qualitative correspondence with the non-linear experiments.

3. **Quantification of the applicability range and limiting conditions:** Targeting MNIST/Fashion-MNIST (FC-VAE), Conv-VAE, and CIFAR-10/SVHN, we empirically demonstrate the reproducibility of AU values and TwoNN ID, the grid and seed dependence of m^{knee} , and that “ $m \gg \hat{d}_{\text{ID}}$ is the dominant condition for TwoNN stability, more important than isometry improvement” (Exp-Real-Geom, Proposition 4). Furthermore, for Conv-VAEs and natural images, we quantify the boundary conditions under which $\text{AU}/m^{\text{knee}}$ diagnostics become difficult, as five operational criteria for stable identification (Sections 6.6 and 6.7).

1.3. Two Kinds of “Dimension”: Local Intrinsic Dimension and Topological Embedding Dimension

In this paper, we distinguish two notions of manifold “dimension”: d_{ID} (the local intrinsic dimension, the target of TwoNN/MLE estimation) and d_{emb} (the topological minimal embedding dimension, the minimal dimension required for an embedding without self-intersection) (for the Swiss Roll, $d_{\text{emb}} = d_{\text{ID}} = 2$; for the Torus, $d_{\text{emb}} = 3 > d_{\text{ID}} = 2$).

Note, however, that **the main contribution of this paper is not to prove a correspondence with d_{ID} or d_{emb} , but to organize the robustness, limits, and applicability conditions of the diagnostic indicators AU values, TwoNN ID, and m^{knee}** . The conditional correspondence between d_{emb} and AU_{sat} (for the Torus, Möbius strip, etc.) is positioned as an incidental supporting observation (Section 6.5).

1.4. Research Questions

- RQ1** Under what grid-density, seed-count, and architecture conditions can the change point m^{knee} obtained from the AU dynamics of a VAE be used as a diagnostic auxiliary quantity consistent with TwoNN/MLE ID estimates?
- RQ2** Does a framework that interprets the effective latent dimensionality of AEs/VAEs with multiple indicators (AU values, TwoNN/MLE, the MSE elbow, and PCA) provide complementary information relative to any single indicator?
- RQ3** How do bottleneck headroom (the magnitude of $m/\hat{d}_{\text{ID}}$) and geometric regularization (IsometricAE) affect the stability of TwoNN ID estimation?

1.5. Paper Organization

Explicit scope of validation: The validation in this paper covers synthetic manifolds ($d_{\text{true}} \leq 3$), MNIST and Fashion-MNIST (fully connected AE/VAE), Conv-AE/VAE (experiments Exp-Conv-AE and Exp-Conv-Fix), KL annealing (experiment Exp-Conv-Ann), and natural images (CIFAR-10 and SVHN; experiments Exp-Nat-Cifar and Exp-Nat-SVHN). In the natural-image experiments (Exp-Nat-Cifar and Exp-Nat-SVHN), we confirm that the TwoNN auxiliary estimate yields values consistent with those reported by Pope et al. [16], and we further quantify the validity range and limiting conditions of $\text{AU}/m^{\text{knee}}$ diagnostics for Conv-VAEs (we do not claim “generalization” of the entire framework to natural images).

The paper is organized as follows: Section 2 (related work), Section 3 (framework), Section 4 (theoretical background, including Theorem 1), Section 5 (experiments), Section 6

(discussion), and Section 7 (conclusions). Detailed proofs and sensitivity analyses are collected in the appendices.

2. Related Work

2.1. Representation Learning and Autoencoders

Bengio et al. [7] organized the theoretical framework of representation learning and discussed the mechanism by which AEs learn the low-dimensional structure of high-dimensional data based on the manifold hypothesis [6]. β -VAE [20] promotes disentangled representations by increasing the KL-regularization weight β , which directly affects AU. Lucas et al. [21] analyzed, within a linear-VAE framework, the conditions under which posterior collapse (a substantial reduction of AU) occurs in ELBO minimization, and showed theoretically that AU saturates near d_{ID} . From the perspective of score matching, the DAE analysis [23] showed that the reconstruction directions of denoising autoencoders selectively suppress the normal directions of the manifold. The CAE [22] learns contractive representations by regularizing the encoder-side Jacobian.

2.2. Intrinsic-Dimension Analysis of Deep Representations

Pope et al. [16] showed via TwoNN estimation that the intrinsic dimension of natural images is far lower than the pixel dimension (CIFAR-10: $d_{\text{ID}} \approx 26$ at 32×32). Ansuini et al. [17] quantified the patterns by which ID changes across the layers of deep networks, documenting contraction and expansion phenomena of the representations. Valeriani et al. [19] systematically analyzed the geometric structure of the hidden representations of large transformer models. Birdal et al. [18] analyzed the relationship to generalization error by integrating persistent homology with ID. Our study is complementary to these works that “measure the ID of existing networks,” and differs in that it tracks the formation of effective dimensionality at the **design stage** and during the **training dynamics** of AEs/VAEs.

As intrinsic-dimension estimation methods, we adopt TwoNN [8] (two-nearest-neighbor distance ratios) and MLE [9] (maximum-likelihood estimation). DANCo [15] is more precise but computationally more expensive. While UMAP [25] is widely used as a non-linear dimensionality-reduction technique for visualization, our study focuses not on visualization but on the **quantitative interpretation** of the bottleneck dimension.

2.3. Active Units and Posterior Collapse

The active-unit (AU) count [20,21] is the number of latent dimensions whose posterior-mean variance exceeds a threshold δ , and it quantifies the number of dimensions in which KL regularization is functioning effectively. Lucas et al. [21] showed, within a linear-VAE framework, that ELBO minimization induces posterior collapse and that AU saturates near d_{ID} . In contrast, our Theorem 1 **quantifies Lucas et al.’s qualitative argument as the closed-form threshold** $\lambda_j > \beta\tau^2$ in terms of the data eigenvalues λ_j , the noise τ^2 , and the regularization β ; its novelty lies in carrying out a quantitative derivation under the linear-Gaussian rate–distortion surrogate, using an explicit connection to Shannon’s water-filling theorem and the KKT conditions (it also subsumes the $\beta = 1$ special case of PPCA by Tipping & Bishop [5]). Moreover, the systematic analysis of AU growth dynamics on real data—in particular the concept of m^{knee} and its correspondence with effective representational dimensionality, its limiting conditions, and stabilization strategies—is systematically validated in this study (Theorem 1, Proposition 2, Section 6.2).

2.4. Geometric Autoencoders

Nazari et al. [14] proposed geometric autoencoders, introducing a regularizer that makes the encoder preserve the geometric structure of the manifold. Kato et al. [13] proposed IsometricAE, which realizes an isometric embedding by driving the pullback metric of the decoder Jacobian toward the identity matrix. In this study, we position IsometricAE as “an auxiliary tool that improves a precondition (isometry) for the readout accuracy of effective dimensionality” (RQ3, experiment Exp-Syn-Iso).

3. Framework: Reading Out the Effective Latent Dimensionality

3.1. Problem Setup and Main Notation

We define an AE/VAE with input dimension N and bottleneck dimension m : an encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and a decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. For a VAE, the encoder outputs a posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$, and the KL regularization (with strength β) acts as an information bottleneck.

Definition of Active Units (AU):

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2} \quad (1)$$

where $\mu_k(x)$ is the k -th component of the posterior mean and δ is a threshold (following Lucas et al. [21]). AU is extremely robust to this threshold choice: a sensitivity analysis confirms that the AU values remain essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (see Appendix G).

Definition of the AU increase rate (two steps): We first define the raw increase rate $s(m_i)$ on the grid as

$$s(m_i) = \frac{\text{AU}(m_i) - \text{AU}(m_{i-1})}{m_i - m_{i-1}} \quad (2)$$

where m_{i-1} is the immediately preceding value on the grid. We then define the normalized increase rate $\rho(m_i)$ as

$$\rho(m_i) = \frac{s(m_i)}{\max_j s(m_j)} \quad (3)$$

so that $\rho(m_i) \in [0, 1]$.

Table 1 lists the **main notation**.

Table 1. Notation used throughout this paper

Symbol	Meaning	Section
m	Bottleneck dimension (design parameter)	§3
$m^{\text{cand}} (= m^{\text{cand}})$	Diagnostic candidate dimension (not a unique optimum)	§3
$m^{\text{sat}} (= m^{\text{sat}})$	AU sharp saturation point (diagnostic candidate for synthetic manifolds)	§3
$m^{\text{knee}} (= m^{\text{knee}})$	AU slowdown point (rough heuristic for real data)	Equation (4)
AU_{sat}	Maximum AU value over the grid	Equation (1)
$\hat{d}_{\text{ID}}$	ID estimate via TwoNN/MLE	§3
d_{true}	True intrinsic dimension of synthetic manifold (known)	§5.2
d_{emb}	Topological minimum embedding dimension ($d_{\text{emb}} \geq d_{\text{ID}}$)	§1.3
κ	Decoder Jacobian condition number (isometry indicator)	§2.4
$s(m_i)$	Raw AU increase rate per unit dimension	Equation (2)
$\rho(m_i)$	Normalized AU increase rate ($\in [0, 1]$)	Equation (3)
τ_{knee}	Slowdown detection threshold (default: 0.25)	Equation (4)

The core of our framework is the explicit classification of each indicator as a **robust indicator** or an **exploratory indicator**, followed by their complementary use (Table 2). This classification is an empirical one grounded in the experimental findings reported

later (§5.3.5), and it forms the core of a design philosophy that systematically avoids the misdiagnoses caused by overreliance on any single indicator.

On the meaning of “robust indicator” and “exploratory indicator”: “Robust” here does not mean unbiased with respect to the true intrinsic dimension. We use the term in the sense that, at least under our experimental conditions, the variance with respect to random seeds is small. “Exploratory,” in contrast, means that the indicator is useful for grid design and hypothesis generation but is not treated as a unique estimate of the effective latent dimensionality.

Table 2. Classification and empirical reliability of diagnostic indicators

Indicator	Category	Empirical reliability (inter-seed σ)	Primary use
AU values (AU(m))	Robust indicator	$\sigma \leq 0.5$ (MNIST 300ep)	Direct observation of effective dimensionality; independent of reference AE geometry (depends on β , δ , model capacity)
TwoNN ID ($\hat{d}_{\text{ID}}$)	Numerically stable auxiliary reference[†]	$\sigma \leq 0.12$ ($m = 64$)	Reference estimate of intrinsic dimension
MSE elbow	Auxiliary indicator	Qualitative only	Reconstruction-side verification
PCA cumulative variance m^{knee} (AU slowdown point)	Auxiliary indicator Exploratory indicator	Linear assumption only $\sigma = 15$ (sparse grid); $\sigma = 0.9$ (dense grid)	Linear baseline Rough heuristic of effective dimensionality

[†] TwoNN ID is

a **supplementary reference estimator** subject to isometry bias ($\kappa \approx 10^9$); see §3.2 for details. σ of m^{knee} is for sparse grid; improved to $\sigma = 0.9$ with dense grid (Exp-Mnist-Dense), where grid design is a major factor (§5.3.7).

Remark 1 (Terminology). We distinguish the following terms throughout this paper. The **effective latent dimensionality** is the umbrella term for the number of bottleneck dimensions that the AE/VAE actually exploits—the quantity the designer wishes to know. $\hat{d}_{\text{ID}}$ (estimated intrinsic dimension) denotes the point estimate obtained by TwoNN/MLE (via a reference AE, subject to isometry bias). m^{knee} denotes the observational indicator of effective dimensionality read from the AU dynamics of the VAE. AU_{sat} denotes the maximum AU value over the grid (the saturation value for synthetic manifolds; a reference value for real data). All of these are estimates of the “effective latent dimensionality” obtained through different approximation and observation procedures, and in principle they can be expected to agree (an empirical confirmation under the conditions of this study; §5).

3.2. Pipeline for Reading Out the Effective Latent Dimensionality

The framework proposed in this study is an interpretation pipeline that “reads out the effective latent dimensionality learned by an AE/VAE in two stages” (Figure 1).

Stage 1: Approximate estimation of the intrinsic dimension (auxiliary role)

In this paper, we estimate the intrinsic dimension $\hat{d}_{\text{ID}}$ in two ways: (a) for **synthetic manifolds**, TwoNN/MLE is applied directly in the input space (allowing comparison with d_{true}); (b) for **real data (MNIST, etc.)**, to avoid the curse of high dimensionality, the estimate is obtained via the latent representation of a larger reference AE ($m_{\text{ref}} = 256$).

Explicit statement of limitations: The latent space of the reference AE in method (b) ($m = 256 \gg \hat{d}_{\text{ID}}$) does not guarantee isometry (auxiliary measurement: $\kappa \approx 10^9$; §6.7), so the resulting $\hat{d}_{\text{ID}}$ is an approximation subject to isometry bias. In this paper, we treat $\hat{d}_{\text{ID}}$ as **auxiliary prior information** indicating the candidate range for m^{knee} , and by design we entrust the primary readout of the effective dimensionality to the AU dynamics of Stage 2. This avoids circular reasoning stemming from the non-isometry of the reference AE. We

ensure practical stability by confirming that the estimates converge across multiple values of m (e.g., $\{64, 128, 256\}$).

Stage 2: Reading out the effective dimensionality via AU dynamics

We sweep the bottleneck dimension m over the discrete grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$, training a VAE ($\beta = 4$) at each m and computing $AU(m)$. Depending on the AU pattern, we read the diagnostic candidate dimension m^{cand} ($= m^{\text{cand}}$) in one of two ways (m^{cand} is not the unique “optimal” bottleneck dimension but a diagnostic candidate dimension based on multiple indicators):

Saturation type (simple data such as synthetic manifolds) When AU saturates sharply at a specific m , we use that saturation point m^{sat} as the diagnostic candidate dimension $m^{\text{cand}} = m^{\text{sat}}$ (since AU values are robust, this is quite reliable under these conditions).

Gradual-increase type (real data, e.g., MNIST) When AU increases gradually with m , we refer to the slowdown point m^{knee} as a rough design guideline (an exploratory indicator):

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\} \quad (4)$$

with $\tau_{\text{knee}} = 0.25$ (default; see Appendix C). m^{knee} roughly captures “the point at which the VAE first performs significant dimensional compression,” but its value depends on the m grid and on the random seed, so the final judgment is made by cross-checking against the AU values and the TwoNN ID (§6.1). Even if m^{knee} and $\hat{d}_{\text{ID}}$ do not agree, the framework as a whole is not invalidated.

CKA [24] and the reconstruction-error (MSE) elbow serve as auxiliary criteria, and the overall judgment is made from multiple indicators.

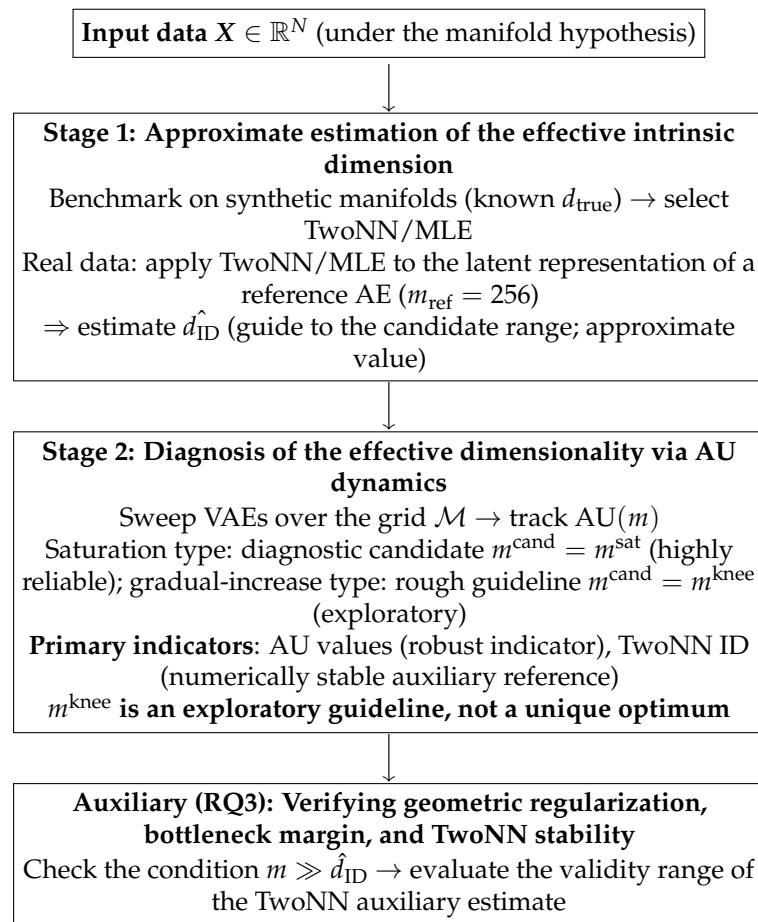


Figure 1. Diagnostic pipeline for effective latent dimensionality. The diagnostic candidate dimension m^{cand} is not a unique optimal solution but a candidate based on multiple indicators (AU values, TwoNN ID, m^{knee}).

3.3. Interpretive Significance of m^{knee}

We now describe the mechanism by which m^{knee} functions as an indicator of the effective representation dimensionality.

In the course of minimizing the VAE's ELBO, the KL regularization (the information bottleneck term) deactivates surplus latent dimensions that are not needed for reconstruction (i.e., it induces posterior collapse) [20,21]. In the regime $m < \hat{d}_{\text{ID}}$, the bottleneck impedes reconstruction, so the VAE activates all available dimensions as AU ($\text{AU}(m) \approx m$). Once m exceeds roughly $\hat{d}_{\text{ID}}$, surplus dimensions begin to appear and the KL regularization exerts pressure to deactivate them, so the AU increase rate slows down. This slowdown point m^{knee} tends to agree with the data's effective intrinsic dimension $\hat{d}_{\text{ID}}$ under specific conditions (e.g., Fashion-MNIST), but its numerical reproducibility depends on the m grid and on random seeds and is not guaranteed (§5.3.5, §6.1). As a framework, regardless of whether m^{knee} agrees, the readout of the effective dimensionality through the combination of AU values (a robust, directly observed quantity) and the TwoNN ID (a numerically stable auxiliary reference estimator) constitutes the central insight of this framework.

4. Theoretical Background

In this section, we present the theoretical foundations that support the framework. Proposition 1 is a lower bound based on standard topological facts about topological dimension and embedding dimension; its rigorous claim is restricted to the idealized premise of "the dimension required to achieve zero reconstruction error." Theorem 1 is an analytical result under the linear-Gaussian rate–distortion surrogate and gives the AU

activation threshold in closed form (proof details in A). Proposition 2 is a lemma under an idealized monotone-saturation model; note that in practical non-linear settings, errors exceeding the upper bound can occur. Proposition 4 is not a guarantee of unbiasedness but a sketch of sufficient conditions for conditional stability. Proposition 3 follows directly from the definition of isometry.

Proposition 1 (Topological Lower Bound on the Bottleneck Dimension). *For a smooth d -dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$ to be reconstructed by a continuous AE without self-intersection, ideally achieving zero reconstruction error, a bottleneck dimension $m \geq d$ is necessary. (Practical settings involving finite samples, finite capacity, and approximate reconstruction are outside the scope of this idealization.)*

Proof. Zero reconstruction error requires $g \circ f|_{\mathcal{M}}$ to be close to the identity map, so $f|_{\mathcal{M}} : \mathcal{M} \rightarrow \mathbb{R}^m$ must be a continuous injection. By standard properties of the topological embedding dimension (Hatcher [10], Chapter 2), no continuous injection exists from a d -dimensional manifold into a Euclidean space of dimension $m < d$; hence $m \geq d$ is a necessary condition. $\square$

This proposition justifies $\hat{d}_{\text{ID}}$ as a reference point for a principled lower bound on the bottleneck dimension (in practice, we use it as a "candidate range for the effective dimensionality"). Experiment Exp-Syn-1 confirms this: at the transition $m = 1 \rightarrow 2$, the MSE improves by roughly a factor of 35 on the Swiss Roll (Section 5.2).

Remark 2 (Differences from Prior Work). *Lucas et al. [21] qualitatively discussed the relation between posterior collapse and intrinsic dimension in linear VAEs. Probabilistic PCA (PPCA) by Tipping & Bishop [5] derives the activation condition of principal components under isotropic noise $\tau^2 I$, which corresponds to the special case $\beta = 1$. The novelty of our theorem is that, under the linear-Gaussian rate-distortion surrogate, it derives the threshold $\lambda_j > \beta\tau^2$ for the β -ELBO with $\beta \neq 1$ by explicitly connecting it to Shannon's water-filling theorem [1], and provides the closed-form upper bound on the number of active units, $d^*(\beta) = |\{j : \lambda_j > \beta\tau^2\}|$, together with monotone saturation. This yields a quantitative and actionable design guideline: the product of β and τ^2 acts as an effective SNR threshold.*

Theorem 1 (AU Activation Threshold under the Linear-Gaussian Rate-Distortion Surrogate). *For a β -VAE with linear encoder, linear decoder, and Gaussian prior (detailed setting in A), under the linear-Gaussian rate-distortion surrogate, the water-filling theorem and the KKT conditions yield the following:*

1. **(Activation threshold)** *The j -th eigen-direction becomes active iff*

$$\lambda_j > \beta\tau^2 \quad (5)$$

where λ_j is the j -th eigenvalue of the data covariance and τ^2 is the decoder noise variance.

2. **(Upper bound on the optimal number of active eigen-directions)** $A^*(m) \leq \min(m, d^*(\beta))$
 $d^*(\beta) := |\{j : \lambda_j > \beta\tau^2\}|$
where $A^(m)$ is the number of active eigen-directions at the global optimum.*
3. **(Monotone saturation at the optimum)** $A^*(m)$ is monotone non-decreasing in m and saturates for $m \geq d^*(\beta)$ (i.e., $A^*(m) = d^*(\beta)$).

A detailed derivation is given in B. The quantity $\beta\tau^2$ acts as the "water level" of water-filling: only eigen-directions whose eigenvalues exceed it become active.

Remark 3 (Relation to Empirical AU). Item (3) of Theorem 1 establishes the monotonicity of the number of active eigen-directions $A^*(m)$ at the global optimum. In contrast, practical non-linear VAEs are trained independently for each m by stochastic optimization, so the observed empirical $AU(m)$ can be **non-monotone** due to finite-sample effects, initialization dependence, and optimization noise (e.g., in Table 9, $AU=26$ at $m=128$ falls below $AU=29$ at $m=64$). Therefore, the empirical m^{knee} must be treated not as a rigorous estimate of the saturation point of A^* , but as an exploratory diagnostic indicator derived from the shape of the AU curve.

Proof sketch (details in A). Analyzing the β -ELBO contribution of each PCA direction j independently (a standard result for linear-Gaussian models), the ELBO gain ΔL_j from activating direction j is positive only when $\lambda_j > \beta\tau^2$ (the correspondence with Shannon's water-filling theorem). By the independence across dimensions, the relation between the optimal AU count and $d^*(\beta)$, as well as the monotone saturation, follows. $\square$

Remark 4. Key implications of Theorem 1: (i) The larger β or τ^2 , the higher the saturation threshold $\beta\tau^2$, and the fewer the active dimensions (stronger dimensional compression). (ii) A vanilla AE without KL regularization corresponds, from the viewpoint of this surrogate, to the limit $\beta=0$: the threshold $\beta\tau^2$ vanishes, so directions with positive variance are hardly suppressed. Consequently, the number of utilized latent coordinates—the AE counterpart of the VAE's AU—can be interpreted as approaching m (consistent with experiment Exp-Syn-AU; Section 5.2). (iii) Although the theorem holds in the linear-Gaussian setting, it provides, as a qualitative extension to real data, the rule of thumb that "dimensions whose effective SNR falls below β are compressed."

Proposition 2 (Grid-Resolution Dependence of m^{knee} under the Idealized Monotone-Saturation Model). Assume the idealized model in which $AU(m)$ saturates ideally and monotonically at $d^*(\beta)$ (changing in a step-like manner, with no stochastic noise). Then, for $m^{\text{knee}}(\mathcal{G})$ on a grid $\mathcal{G} = \{m_1 < \dots < m_K\}$,

$$|m^{\text{knee}}(\mathcal{G}) - d^*(\beta)| \leq \max_i (m_{i+1} - m_i) \quad (6)$$

holds (the error is bounded above by the maximum grid spacing).

Caution: in practical non-linear VAEs this idealization does not hold, and shape noise in $AU(m)$ due to stochastic optimization can produce errors that substantially exceed the upper bound in Equation (6) (consistent with the observations in Section 5.3.5). This proposition should be treated not as a general instability theorem but as a lemma providing guidance for grid design.

Proof. In the idealized monotone-saturation model, $AU(m)$ changes step-wise at $d^*(\beta)$, so if $d^*(\beta) \in [m_{i_0}, m_{i_0+1})$, then m^{knee} is detected as m_{i_0+1} (or m_{i_0}). The error is $\leq m_{i_0+1} - m_{i_0} \leq \max_i (m_{i+1} - m_i)$. $\square$

Remark 5. Experimental implication of this proposition: the grid $\{4, 8, 12, 16, 20, 32, 64\}$ contains a maximum spacing of 32 (from 32 to 64), so for $d^* \approx 10$ a large error with $m^{\text{knee}} \in \{32, 64\}$ can arise (consistent with Section 5.3.5). A dense grid combined with a multi-seed ensemble is a practical stabilization strategy (Section 6.2).

Proposition 3 (Pullback Metric and Local Isometry). A necessary and sufficient condition for the decoder $g: \mathbb{R}^m \rightarrow \mathbb{R}^N$ to be locally isometric in a neighborhood of a point z is that the pullback metric satisfies $G(z) = J_g(z)^\top J_g(z) = I_m$. That is, $\|J_g(z)dz\|^2 = \|dz\|^2$ holding for every infinitesimal displacement dz is equivalent to $G(z) = I_m$ (all singular values equal to 1).

By contrast, the condition number $\kappa(z) = \sigma_{\max}(J_g)/\sigma_{\min}(J_g)$ is an indicator of the anisotropy of the local map: $\kappa = 1$ means that all singular values are **equal** (isotropic scaling), but does not

guarantee that this scale equals 1. For example, if J_g magnifies all directions by a factor of 2, then $\kappa = 1$ but the map is not isometric. Accordingly, in this paper we use κ not as isometry itself but as an auxiliary indicator of local anisotropy.

Proof. By the definition of isometry, $\|J_g dz\|^2 = \|dz\|^2$ for every $dz \in \mathbb{R}^m$ is equivalent to $J_g^\top J_g = I_m$ (all singular values equal to 1). Since $\kappa = 1$ only means that all singular values are equal, whether their value is 1 is a separate matter; hence $\kappa = 1$ is not a necessary and sufficient condition for isometry (see the example above). $\square$

This proposition explains why an isometry bias arises in ID estimation using a reference AE (Section 6.7) and motivates the IsometricAE (experiment Exp-Syn-Iso).

The remaining interpretive propositions (effective rank of the decoder Jacobian, noise-direction selectivity, and neighborhood-structure preservation) are given in A.

4.1. Qualitative Correspondence between Linear Theory and Non-Linear Experiments

Theorem 1 (the water-filling principle) holds under the linear-Gaussian rate-distortion surrogate. In non-linear settings, a powerful decoder achieves an effective noise variance $\tau_{\text{eff}}^2 \ll 1$, so $d^*(\beta) = |\{j : \lambda_j > \beta \tau_{\text{eff}}^2\}|$ becomes substantially larger than the prediction under the linear assumption ($\tau^2 = 1$). Moreover, the smooth AU growth curves produced by non-linear optimization replace the step-like saturation transition of the linear setting, causing the instability of m^{knee} on sparse grids.

We demonstrate this qualitative inheritance (AU monotonicity, slowdown of growth beyond the effective dimensionality) and the quantitative divergence through a MNIST PCA spectrum analysis (experiment Exp-Mnist-PCA) in Section 5.3.6.

4.2. Conditional Stability of TwoNN: Theoretical Formulation

In this section, motivated by the experimental observations of Section 5.2.4 (Exp-Stab, Exp-Geom, and Exp-Iso-ID), we theoretically support, as an explanatory proposition, the question "under what conditions can TwoNN be used as an auxiliary reference estimator?" **This proposition does not guarantee the unbiasedness of TwoNN; it is an interpretive proposition giving a sketch of sufficient conditions for conditional stability** (positioned not as a mathematically rigorous theorem but as a theoretical rationale for a design guideline).

Proposition 4 (Conditional Stability of TwoNN). *Suppose the encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ is locally bi-Lipschitz, i.e., in a neighborhood U of every point z ,*

$$(1 - \varepsilon)\|v\| \leq \|J_f \cdot v\| \leq (1 + \varepsilon)\|v\|, \quad \varepsilon \in [0, 1] \quad (7)$$

holds, and suppose further that nearest-neighbor ranks are preserved with high probability (Trustworthiness $\geq 1 - \delta_T$ with small δ_T). Then the change in the empirical distribution of the two-nearest-neighbor distance ratio r_2/r_1 , measured by the earth mover's distance (EMD), is bounded as

$$\text{EMD}(P_{r_2/r_1}^{(\text{input})}, P_{r_2/r_1}^{(\text{latent})}) \leq C(\varepsilon, \delta_T) \quad (8)$$

and consequently the deviation of the TwoNN estimate, $|\hat{d}_{\text{ID}}^{(\text{latent})} - \hat{d}_{\text{ID}}^{(\text{input})}|$, is also bounded above by a constant governed by $C(\varepsilon, \delta_T)$. ($C(\varepsilon, \delta_T) \rightarrow 0$ as $\varepsilon \rightarrow 0$ and $\delta_T \rightarrow 0$.)

Proof. *Proof sketch:* By Equation (7), the change factor of distances $d(x_i, x_j)$ lies within the range $(1 - \varepsilon)^2$ to $(1 + \varepsilon)^2$. Since the distance ratio r_2/r_1 is a ratio rather than an absolute distance, uniform scale deformations (global rescaling) cancel out, but the non-isometric component with $\varepsilon \neq 0$ alters the ratio. Trustworthiness $\geq 1 - \delta_T$ guarantees

that "the two nearest neighbors in latent space coincide with those in input space with high probability," probabilistically suppressing large fluctuations of r_2/r_1 caused by rank reversals. Under these two conditions, the change in the distribution of r_2/r_1 is bounded above by a continuous function $C(\varepsilon, \delta_T)$ of ε and δ_T ($C = 0$ when $\varepsilon = 0$ and $\delta_T = 0$, coinciding with the isometric, perfectly neighbor-preserving case). (This is a finite-sample, finite-dimensional approximation; a rigorous bound depends on the specific setting of the isometry constants.) $\square$

Remark 6 (Experimental Implications of Proposition 4). This proposition suggests that "isometry $\varepsilon \approx 0$ is the sole condition for TwoNN accuracy" is not the case; rather, "the improvement of Trustworthiness (preservation of neighbor ranks) induced by $m \gg d_{\text{ID}}$ may be the dominant factor for stability." In experiment Exp-Stab (Section 5.2.4), the difference in TwoNN ID remained within $\pm 2.3\%$ over the range $\kappa \in [1.06, 2.16]$, whereas TwoNN recovered upon moving from $m = d_{\text{true}}$ to $m > d_{\text{true}}$ —consistent with the decrease in δ_T (the neighbor-reversal rate) being more dominant than the change in ε (isometry). This proposition does not claim that TwoNN is an "unbiased estimator of the true d_{ID} "; it theoretically supports "under what conditions TwoNN can be operated as an auxiliary reference estimator." Experiment Exp-Real-Geom (Section 5.10) directly validates this implication on real data.

5. Experiments

Three-tier classification of experiments: The experiments in this paper are clearly classified into the following three tiers (corresponding to the "Category" column of Table 3).

Main experiments Empirical validation of the framework core: Exp-Syn-1, Exp-Syn-AU, Exp-Mnist-Base, Exp-Mnist-Multi, Exp-Mnist-Dense, Exp-Fashion, Exp-Real-Geom, and Exp-Nat-Cifar. Their primary validation targets are the robustness of AU values and TwoNN ID (RQ2) and the stabilization conditions of m^{knee} (RQ1).

Supplementary experiments Verification of prerequisites and interpretive claims of the main experiments: Exp-Stab, Exp-Geom, Exp-Iso-ID, Exp-Syn-Iso, Exp-Conv-AE, Exp-Syn-Top, Exp-Mnist-PCA, Exp-Mnist-Dyn, and Exp-Mnist-Ent. These verify TwoNN numerical stability, IsometricAE isometry, the theory–experiment bridge, and related points (supporting evidence for the main claims).

Boundary experiments Quantification of the applicable range and boundary conditions: Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256, Exp-Conv-M512, Exp-ConvV-HiBeta, Exp-NatImg-AU, Exp-NatImg-Sweep, and Exp-NatImg-Knee. These quantify the conditions under which stable identification of $\text{AU}/m^{\text{knee}}$ becomes difficult for Conv-VAEs and natural images, thereby establishing the validity range and the applicability boundary of the framework.

5.1. Experimental Setup

Common settings: Adam ($\text{lr} = 10^{-3}$), batch size 128, ReLU activations (Sigmoid output layer), random seed 42, PyTorch 2.0.

Synthetic manifolds: hidden layers [64, 32], 200–300 epochs, VAE $\beta = 4.0$, $n = 5,000$ (4,000 training / 1,000 test).

MNIST / Fashion-MNIST: input dimension 784, hidden layers [512, 256], VAE $\beta = 4.0$, grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$, batch size 128. MNIST is trained for 300 epochs ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$), and Fashion-MNIST for 100 epochs ($n_{\text{train}} = 10,000$, $n_{\text{test}} = 2,000$).

ID estimation: TwoNN ($k = 2$; by the definition of the algorithm, $k = 2$ is the only possible choice) and MLE ($k = 10$; near the center of the range $k = 5$ –20 recommended by Levina & Bickel [9], with stability also confirmed for $k = 5, 10, 20$) [8,9]. AU threshold

$\delta = 10^{-2}$ [21] (AU is extremely robust to this threshold choice: a sensitivity analysis confirmed that AU values remain essentially unchanged as δ varies over $[10^{-3}, 10^{-1}]$; Appendix G).

Table 3. Summary of experiment groups. Detailed experiment list in Appendix A. Categories: main experiments (core framework validation), supplementary experiments (prerequisite verification), boundary experiments (validity range quantification).

Experiment Group	Objective	Main Data	Category
Synthetic manifold experiments (Exp-Syn-1, Exp-Syn-AU, Exp-Syn-Iso, Exp-Syn-Top)	Principle verification (MSE elbow, AU saturation, isometry)	Swiss Roll, Torus, Möbius, etc.	Main + Supplementary
FC-AE/VAE experiments (Exp-Mnist-Base, Exp-Mnist-Multi, Exp-Mnist-Dense, Exp-Fashion, Exp-Stab, Exp-Geom, Exp-Iso-ID, Exp-Real-Geom)	Core validation of the multi-indicator diagnostic framework (robustness of AU values and TwoNN ID, m^{knee} stabilization, TwoNN applicability conditions)	MNIST, Fashion- (FC-AE/VAE)	Main
Conv-VAE experiments (Exp-Conv-AE, Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256, Exp-Conv-M512, Exp-ConvV-HiBeta)	Identifying the cause of AU saturation delay in Conv-VAEs and determining the validity range	MNIST (Conv-AE/VAE)	Supplementary + Boundary
Natural image experiments (Exp-Nat-Cifar, Exp-Nat-SVHN, Exp-NatImg-AU, Exp-NatImg-Sweep, Exp-NatImg-Knee)	Validation of external validity of TwoNN auxiliary estimates; quantification of applicability boundary in natural image settings	CIFAR-10, SVHN	Supplementary + Boundary

5.2. Synthetic Manifold Experiments: Principle Verification (RQ1, RQ2, RQ3)

Using synthetic manifolds with known d_{true} , we verify the fundamental validity of the proposed framework.

5.2.1. Correspondence Between MSE Elbows and Intrinsic Dimension (Experiment Exp-Syn-1)

For the Swiss Roll ($d_{\text{true}} = 2$, simply connected) and the Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$), we confirmed that the AE reconstruction MSE improves sharply at $m = d_{\text{true}}$ (Swiss Roll) or $m = d_{\text{emb}}$ (Torus) (Table 4).

Table 4. Exp-Syn-1: MSE of AE on synthetic manifolds (mean±std over 3 seeds).

m	Swiss Roll MSE	Torus MSE
1	1.016 ± 0.057	0.388 ± 0.017
2	$0.029 \pm 0.002 \leftarrow$ elbow	0.085 ± 0.004
3	0.0008 ± 0.0001	$0.0005 \pm 0.0000 \leftarrow$ elbow
4	0.041 ± 0.015	0.019 ± 0.002

For the Swiss Roll, the MSE improves by a factor of approximately 35 from $m = 1$ to $m = 2$, experimentally confirming Proposition 1. For the Torus, the elbow appears at $m = 3$ ($= d_{\text{emb}}$), showing that perfect reconstruction is impossible at $d_{\text{ID}} = 2$.

5.2.2. AU Saturation and Topological Structure (Experiments Exp-Syn-AU and Exp-Syn-Top)

We compared the AU pattern of the VAE ($\beta = 4$) with that of a vanilla AE (Table 5). **Caveat:** AU is an indicator of whether the coordinate-wise variance of the VAE posterior mean exceeds a threshold, and it is not directly defined for AEs, which lack KL regular-

ization. The “AU” reported for the AE in the table is an AE-style pseudo-AU indicating whether the encoding variance of each latent coordinate exceeds $\delta = 10^{-2}$; it is an auxiliary observation showing the tendency of all coordinates to be effectively used.

Table 5. Exp-Syn-AU: VAE active units (AU) for Swiss Roll (upper) and Torus (lower) (mean $\pm$ std, 3 seeds).

	m	Vanilla AE (AU)	VAE (AU, $\beta = 4$)
Swiss Roll ($d_{\text{true}} = 2$)	2	2	2.0 ± 0.0 (saturation)
	4	4	2.0 ± 0.0
	8	8	2.0 ± 0.0
	16	16	2.0 ± 0.0
Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$)	4	4	2.0 ± 0.0
	6	6	3.0 ± 0.0 (saturation)
	8	8	3.0 ± 0.0
	16	16	3.0 ± 0.0

Whereas the vanilla AE exhibits $\text{AU} = m$ (no saturation), the VAE saturates at $\text{AU}_{\text{sat}} = 2$ ($= d_{\text{true}}$) for the Swiss Roll and at $\text{AU}_{\text{sat}} = 3$ ($= d_{\text{emb}} > d_{\text{ID}}$) for the Torus (standard deviation 0.0 across all 3 seeds). This is consistent with the qualitative implication of Theorem 1 (the water-filling AU saturation mechanism), and suggests that on the Torus the KL regularization may begin compressing near the topological embedding dimension (an empirical observation under the conditions of this study; note that this finding is preprocessing-dependent and conditional).

Table 6 shows the results of experiment Exp-Syn-Top (correspondence between AU_{sat} and d_{emb} on topological manifolds).

Table 6. Exp-Syn-Top: Topological manifolds and VAE AU_{sat} (standardized conditions; Swiss Roll excluded from the main evidence, see text).

Manifold	d_{ID}	π_1	d_{emb}	AU_{sat} (standardized)
Torus T^2	2	$\mathbb{Z}^2$	3	3 (main result)
Möbius Band	2	$\mathbb{Z}$	3	3 (main result)
Circle S^1	1	$\mathbb{Z}$	2	2 ($m \geq 6$, conditional)

Caveat: The Swiss Roll yields $\text{AU}_{\text{sat}} = 2$ ($= d_{\text{true}}$) in Table 5 (standardized conditions), but $\text{AU}_{\text{sat}} = 3$ is sometimes observed under different preprocessing conditions, indicating a strong preprocessing dependence. We therefore restrict the main evidence supporting the correspondence between AU_{sat} and d_{emb} to the Torus and the Möbius Band, and exclude the Swiss Roll from the evidence for this correspondence. For the Torus and the Möbius Band, $\text{AU}_{\text{sat}} = d_{\text{emb}} = 3$ was observed under the conditions of this study. **However**, this correspondence depends on our experimental conditions (standardized preprocessing, $\beta = 4$, sufficient convergence) and is not guaranteed as a theoretical necessity (see §6).

5.2.3. Achieving Isometry with IsometricAE (Experiment Exp-Syn-Iso)

On the Swiss Roll ($m = 2$), we compared the isometry of four model types (Table 7).

Table 7. Exp-Syn-Iso: Comparison of four models on Swiss Roll ($m = 2$) (mean $\pm$ std, 3 seeds).

Model	MSE	κ	NSR
AE	0.030 ± 0.003	1.251 ± 0.184	0.979 ± 0.036
CAE	0.036 ± 0.008	1.438 ± 0.105	0.659 ± 0.105
IsometricAE	0.029 ± 0.003	1.040 ± 0.009	0.753 ± 0.051
DAE	0.032 ± 0.003	1.179 ± 0.139	0.609 ± 0.085

Only IsometricAE achieves $\kappa \approx 1.04$ (isometry), consistent with Proposition 3, showing that an isometric latent space can be realized without substantially sacrificing MSE. An isometric latent space is a prerequisite that improves the reliability of ID estimation, and it contributes to the theoretical accuracy of Stage 1 ID estimation (with a trade-off against computational cost; §6.7).

5.2.4. Numerical Stability of TwoNN ID and the Effect of κ (Experiments Exp-Stab, Exp-Geom, and Exp-Iso-ID; details in Appendix E)

Three supplementary experiments (the effect of κ on the Swiss Roll, an MNIST m -sweep, and a multi-dimension IsometricAE comparison) yielded a consistent message: **“the dominant factor for TwoNN stability is the condition $m \gg \hat{d}_{ID}$, not the magnitude of κ .”**

In Exp-Stab (Swiss Roll), at $m = 2 = d_{\text{true}}$ both the AE ($\kappa = 1.79$) and the IsometricAE ($\kappa = 1.06$) underestimate similarly with $\text{TwoNN} \approx 1.55$, whereas at $m = 4 > d_{\text{true}}$ TwoNN recovers to ≈ 2.03 (insufficient bottleneck margin is the primary cause; the effect of the κ difference is minor). In Exp-Geom (MNIST AE m -sweep), TwoNN converges stably to 10.16 for $m \geq 48 \approx 5\hat{d}_{ID}$, and even over $m = 64\text{--}128$, where κ jumps from 4.25 to 37.0, TwoNN remains stable at 10.17–10.98 (detailed table in Appendix E, Table A2). In Exp-Iso-ID (Swiss Roll, $m \in \{2, 3, 4, 6, 8\}$), for $m \geq 3$ the differences in TwoNN ID remained within $\pm 2.3\%$ despite κ differences in $[1.1, 2.1]$ (detailed table in Appendix E, Table A3).

The findings of these three experiments reinforce the design rationale for using a reference AE ($m_{\text{ref}} = 256 \gg \hat{d}_{ID} \approx 10$) in Stage 1, and also show that there is a risk of underestimation for $m < 32 \approx 3\hat{d}_{ID}$ (note that “numerically stable” does not mean “geometrically accurate”; §6.7).

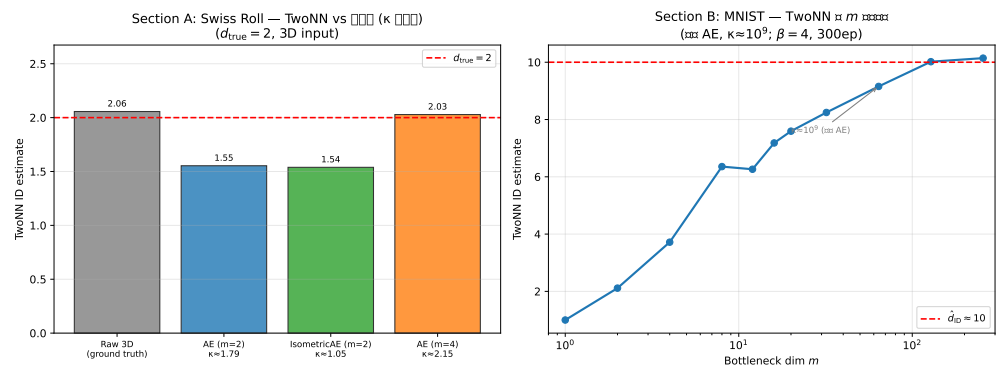


Figure 2. Exp-Stab/Exp-Geom: (Left) TwoNN ID on Swiss Roll — AE and IsometricAE both underestimate at $m = d_{\text{true}}$; TwoNN recovers at $m = 4 > d_{\text{true}}$. (Right) MNIST AE m -sweep: TwoNN stabilizes at $m \geq 48$ despite κ jumping to 37.0 at $m = 128$. Detailed tables in Appendix D.

5.3. MNIST: Reading Off the Effective Representation Dimensionality (Main Experiments, RQ1 and RQ2)

We validate the application of the proposed framework to real data using the MNIST dataset (main experiments). MNIST provides a realistic evaluation environment for the framework in the sense that it is “real data for which d_{true} is unknown but $\hat{d}_{ID}$ can be estimated.”

5.3.1. Intrinsic-Dimension Estimation with an AE (Experiment Exp-Mnist-Base-AE)

Table 8 shows the results for the MNIST AE (300 epochs).

Table 8. Exp-Mnist-Base: MNIST AE results (300 epochs, $n_{\text{train}} = 20,000$).

m	MSE	TwoNN ID	MLE ID	CKA
1	0.0437	1.02	1.15	0.102
4	0.0292	3.86	4.29	0.575
8	0.0191	6.08	6.70	0.725
16	0.0121	7.96	8.61	0.812
32	0.0087	8.84	9.89	0.875
64	0.0062	9.54	11.26	0.908
128	0.0061	9.84	11.45	0.921
256	0.0057	10.02	11.85	0.953

For $m \geq 64$, the TwoNN ID (9.54–10.02) and the MLE ID (11.26–11.85) converge, indicating that the effective intrinsic dimension of MNIST lies roughly in the range $\hat{d}_{\text{ID}} \approx 10$ –12.

Comparison with linear PCA: Applying PCA to MNIST ($n = 20,000$) requires 43 dimensions to reach 80% cumulative explained variance and 86 dimensions to reach 90%. These values are 4–8 times larger than the AE-based estimate (TwoNN ≈ 10), demonstrating the nonlinear manifold structure of MNIST (see Figure 3).

5.3.2. VAE AU Dynamics and m^{knee} (Experiment Exp-Mnist-Base-VAE)

Table 9 and Figure 3 show the AU pattern of the MNIST VAE ($\beta = 4$, 300 epochs).

Table 9. Exp-Mnist-Base: MNIST VAE ($\beta = 4$) results (300 epochs). AU increase rate slows at $m = 12$ ($m^{\text{knee}} = 12$).

m	MSE	AU	TwoNN ID	Change in $r(m)$
1	0.0464	1	1.00	—
4	0.0275	4	3.72	normal growth
8	0.0186	8	6.36	normal growth
12	0.0184	8	6.26	sharp drop $\leftarrow m^{\text{knee}} = 12$
16	0.0148	11	7.18	low
20	0.0134	13	7.59	low
32	0.0121	15	8.25	low
64	0.0103	29	9.16	gradual
128	0.0085	26	10.02	gradual
256	0.0072	36	10.14	gradual

At $m = 12$, $\text{AU} = 8$ (the same value as at the preceding $m = 8$), and the AU increase rate $r(m)$ drops sharply ($r(12) \approx 0$). This triggers the threshold $\tau_{\text{knee}} = 0.25$, and $m^{\text{knee}} = 12$ is observed.

Caveat: this value is a preliminary observation obtained with a single seed (seed 42) and a specific extended grid ($m \in \{1, 4, 8, \dots, 256\}$); subsequent multi-seed experiments (Sections 5.3.4 and 5.3.5) showed that it does not reproduce with the same m values and grid. Although under the single-seed, extended-grid condition it happened to lie close to the TwoNN ID range (≈ 10 –12), **in this paper we treat $m^{\text{knee}} = 10.3 \pm 0.9$ from Exp-Mnist-Dense-300 — based on a dense grid, independent initialization, and multiple seeds — as the main stabilized result** (Section 5.3.7).

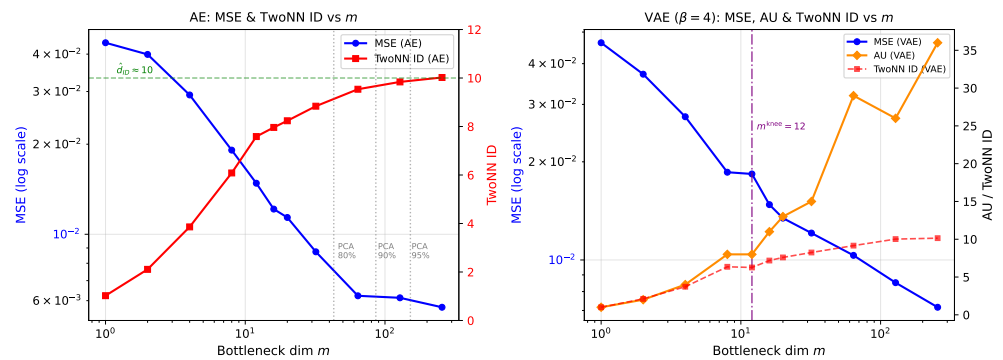


Figure 3. Exp-Mnist-Base: m -dependence of each indicator for MNIST (300 epochs). **Left (AE):** MSE (log scale, left axis) and TwoNN ID (right axis). Vertical dotted lines correspond to PCA cumulative variance 80% (43 dims), 90% (86 dims), 95% (152 dims); the difference from the nonlinear AE estimate (TwoNN ≈ 10) is clear. **Right (VAE, $\beta = 4$):** MSE (log scale, left axis), AU (right axis), TwoNN ID (right axis, dashed). At $m = 12$ (m^{knee}), the AU increase rate drops sharply (arrow), indicating a heuristic slowdown point; the main stabilized result is $m^{\text{knee}} = 10.3 \pm 0.9$ from Exp-Mnist-Dense-300 (Section 5.3.7).

5.3.3. Comparison with Existing Indicators

We compare the proposed indicator m^{knee} with other representative indicators for choosing the bottleneck dimension (Table 10).

Table 10. Comparison of bottleneck dimension estimates on MNIST. m^{knee} entries distinguish sparse-grid and dense-grid conditions.

Method	Estimate	Source	Characteristics / Limitations
PCA (80% var.)	43	Exp-Mnist-Base-AE	Linear; overestimates nonlinear ID
PCA (90% var.)	86	Exp-Mnist-Base-AE	Linear; larger overestimate
MSE elbow (AE)	≈ 16 –20	Exp-Mnist-Base-AE	Intuitive; depends on smoothing
TwoNN ID (ref. AE)	≈ 10 –12	Exp-Mnist-Base-AE	Nonlinear; subject to isometry bias
m^{knee} sparse-grid (3 seeds)	43 ± 15	Exp-Mnist-Multi-300	Exploratory; unreliable at sparse grid
m^{knee} dense-grid (3 seeds)	10.3 ± 0.9	Exp-Mnist-Dense-300	Exploratory; recommended for use

Dense-grid: $m \in \{8, 9, \dots, 16, 18, \dots, 64\}$ with independent initialization per (m, seed) . $m^{\text{knee}} = 10.3 \pm 0.9$ aligns with TwoNN ID (≈ 10). Sparse-grid $m^{\text{knee}} = 43 \pm 15$ is unreliable as a diagnostic reference.

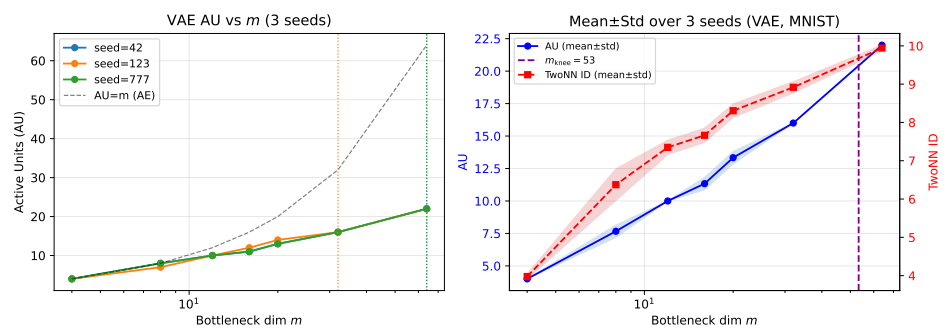
Characteristics of each method: PCA, being a linear method, cannot capture the non-linear manifold structure of MNIST and requires more than four times as many dimensions as the AE-based estimate (TwoNN ≈ 10). The MSE elbow is intuitive, but for a smooth MSE curve (cf. Exp-Mnist-Base) the location of the elbow tends to be ambiguous. TwoNN is effective as a nonlinear ID estimator but is subject to the isometry bias of the reference AE. m^{knee} is complementary to these indicators in that it directly observes the KL dynamics; however, it depends on τ , β , and the m -grid, and is therefore best used within a multi-indicator approach combined with the other indicators rather than in isolation.

5.3.4. Multi-Seed Validation (Experiment Exp-Mnist-Multi-150)

To assess the statistical reliability of m^{knee} , we repeated the MNIST VAE experiment ($\beta = 4$, 150 epochs, $m \in \{4, 8, 12, 16, 20, 32, 64\}$) with three random seeds (42, 123, 777) (experiment Exp-Mnist-Multi-150; Table 11).

Table 11. Exp-Mnist-Multi-150: Mean $\pm$ std of AU and TwoNN ID for 3 seeds (42, 123, 777) at 150 epochs ($\beta = 4$). AU values are highly consistent across seeds, but m^{knee} varies.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.98 ± 0.06	all dims active
8	7.7 ± 0.5	6.37 ± 0.41	
12	10.0 ± 0.0	7.35 ± 0.19	
16	11.3 ± 0.5	7.66 ± 0.18	
20	13.3 ± 0.5	8.31 ± 0.18	
32	16.0 ± 0.0	8.92 ± 0.14	converges to $\hat{d}_{\text{ID}} \approx 10$
64	22.0 ± 0.0	9.94 ± 0.08	
m^{knee} ($\tau = 0.25$): seed 42 = 64, seed 123 = 32, seed 777 = 64 Mean $m^{\text{knee}} = 53 \pm 15$ (150 epochs); the main result is the dense-grid, independently initialized Exp-Mnist-Dense-300: $m^{\text{knee}} = 10.3 \pm 0.9$			

**Figure 4.** Exp-Mnist-Multi: AU patterns of MNIST VAE (3 seeds). Left: AU vs m per seed (dotted lines: m^{knee} per seed); Right: mean $\pm$ std band. AU values are highly consistent across seeds, but m^{knee} varies between 32 and 64.

Key findings: The AU values at each m are highly consistent across the three seeds ($\sigma \leq 0.5$; Table 11). The TwoNN ID also converges to 9.94 ± 0.08 at $m = 64$, confirming that the estimate $\hat{d}_{\text{ID}} \approx 10$ is seed-invariant.

In contrast, m^{knee} ($\tau = 0.25$) varied across seeds: seed 42 = 64, seed 123 = 32, seed 777 = 64 (mean 53 ± 15). This occurs because the AU increase rate $r(m)$ fluctuates subtly around the threshold $\tau = 0.25$ at several m values, and this instability is amplified under the incomplete convergence of 150-epoch training. Note that the extended Exp-Mnist-Base experiment (single seed 42) yielded $m^{\text{knee}} = 12$ at 300 epochs, but the instability persists in the 300-epoch multi-seed experiment described below (Section 5.3.5; mean 43 ± 15). **Although insufficient convergence compounds the problem at 150 epochs, extending training to 300 epochs does not restore the numerical stability of m^{knee} ; the dominant causes turn out to be the dependence on the m -grid and on the random seed (Section 5.3.5).** This finding quantitatively substantiates the limitation stated in Section 6.7.

5.3.5. 300-Epoch Multi-Seed Validation (Experiment Exp-Mnist-Multi-300)

To test whether the m^{knee} instability in the 150-epoch experiment was due to insufficient convergence, we ran an experiment under the same conditions (MNIST VAE, $\beta = 4$, $m \in \{4, 8, 12, 16, 20, 32, 64\}$) with the number of epochs extended to 300 (Exp-Mnist-Multi-300) for three seeds (42, 123, 777) (Table 12).

Table 12. Exp-Mnist-Multi-300: Mean $\pm$ std of AU and TwoNN ID for 3 seeds at 300 epochs ($\beta = 4$). AU and TwoNN ID are highly consistent, but m^{knee} still varies.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 $\pm$ 0.0	3.98 $\pm$ 0.14	all dims active
8	8.0 $\pm$ 0.0	6.67 $\pm$ 0.15	
12	10.0 $\pm$ 0.0	7.41 $\pm$ 0.06	
16	11.7 $\pm$ 0.5	7.95 $\pm$ 0.21	AU still increasing
20	13.3 $\pm$ 0.5	8.37 $\pm$ 0.14	
32	15.7 $\pm$ 0.5	9.07 $\pm$ 0.10	
64	22.3 $\pm$ 0.5	10.03 $\pm$ 0.12	converges to $\hat{d}_{\text{ID}} \approx 10$
$m^{\text{knee}} (\tau = 0.25)$: seed 42 = 64, seed 123 = 32, seed 777 = 32			
Mean $m^{\text{knee}} = 43 \pm 15$ (300 epochs)			

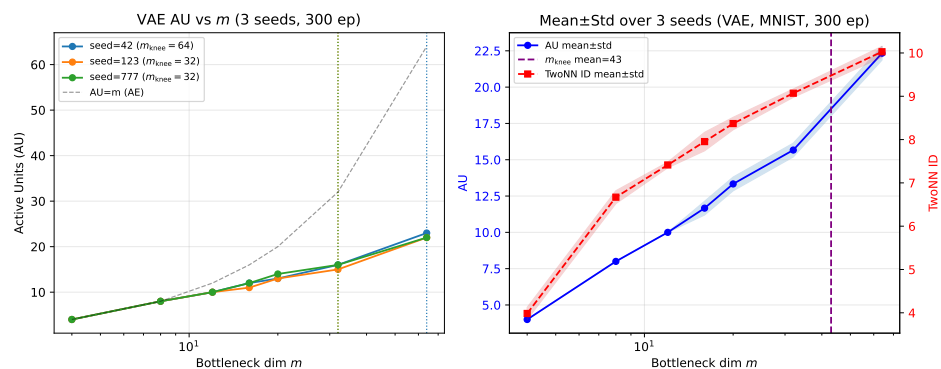


Figure 5. Exp-Mnist-Multi: AU patterns of MNIST VAE (3 seeds, 300 epochs). Left: AU vs m per seed (dotted lines: m^{knee}); Right: mean $\pm$ std band. AU and TwoNN ID bands are very narrow, but m^{knee} varies between 32 and 64 across seeds.

Key findings: The AU values (Table 12) remain highly consistent across the three seeds at 300 epochs ($\sigma \leq 0.5$), and the TwoNN ID is likewise stable at 10.03 ± 0.12 for $m = 64$. We thus confirmed that these quantitative indicators are robust to the seed and to the degree of convergence.

In contrast, m^{knee} still varied between 32 and 64 across seeds even under full 300-epoch convergence (mean 43 ± 15). This is essentially the same variation range as in Exp-Mnist-Multi-150 ($m^{\text{knee}} = 32\text{--}64$), so extending training to 300 epochs did not resolve the instability of m^{knee} .

Note that the $m^{\text{knee}} = 12$ obtained in the extended Exp-Mnist-Base experiment is a preliminary observation that depends on the grid design and the initialization trajectory; in this paper we adopt $m^{\text{knee}} = 10.3 \pm 0.9$ from Exp-Mnist-Dense-300 — obtained under independent initialization, multiple seeds, and a dense grid — as the main result (implementation details in Appendix E). m^{knee} is an exploratory change point derived from the AU growth curve, and its value depends on the experimental setup (grid, seed, and initialization).

5.3.6. Linear Theory–Experiment Bridge (Experiment Exp-Mnist-PCA; Details in Appendix E)

Contrasting the MNIST PCA spectrum ($\lambda_1 = 5.22, \dots, \lambda_{10} = 1.23$) with Theorem 1, we obtain $d^* \approx 10$ for $\tau_{\text{eff}}^2 \approx 0.30$ (back-calculated), qualitatively consistent with the observed TwoNN ID. Whereas the linear setting produces a sharp saturation transition, in nonlinear VAEs gradient competition yields a smooth, gradually increasing curve, which theoretically corroborates the sparse-grid instability of m^{knee} (detailed PCA spectrum tables and theory–experiment comparison tables are given in Appendix E). For Conv-VAEs, the

powerful reconstruction capability of the nonlinear decoder makes τ_{eff}^2 even smaller, so the m required for AU saturation increases substantially relative to the FC-VAE (Section 6.6).

5.3.7. Dense-Grid $\times$ Seed $\times$ Epoch Factor-Isolation Experiment (Experiment Exp-Mnist-Dense)

To rigorously identify which factors cause the instability of m^{knee} , we designed an experiment that isolates three factors: grid density, epochs, and seed. Exp-Mnist-Base and Exp-Mnist-Multi already separate the epoch factor (150 ep vs. 300 ep) and the seed factor (3 seeds). The present experiment, Exp-Mnist-Dense, further isolates the grid-density factor: we trained with $m \in \{8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 24, 28, 32, 40, 48, 64\}$ (17 points, adjacent spacing 1–8) for 300 epochs and 3 seeds. The dense grid covers in particular $m = 8\text{--}16$ (around the estimated $\hat{d}_{\text{ID}} \approx 10$) in steps of 1 (Proposition 2 is an auxiliary upper bound for an idealized model; while a step-1 dense grid may bring m^{knee} close to d^* , AU fluctuations of the nonlinear VAE can also drive the actual error beyond the bound).

Table 13. Exp-Mnist-Dense: Comparison of m^{knee} between dense and sparse grids ($\tau = 0.25$, MNIST FC-VAE, $\beta = 4$). Consistent with the grid-sensitivity bound of Proposition 2.

Condition	Max. grid interval	seed=42	seed=123	seed=777	Mean $\pm$ std
Sparse grid, 150ep (Exp-Mnist-Multi-150)	32	64	32	64	53 $\pm$ 15
Sparse grid, 300ep (Exp-Mnist-Multi-300)	32	64	32	32	43 $\pm$ 15
Dense grid, 300ep (Exp-Mnist-Dense-300)	8 (step 1 for 8–16)	11	9	11	10.3 $\pm$ 0.9

Table 14. Exp-Mnist-Dense: AU(m) for dense grid (seed=42, 300ep, representative values).

m	8	9	10	11	12	13	14	15
AU	8	9	10	10	10	10	12	11
m	16	18	20	24	28	32	48	64
AU	14	13	15	14	15	15	19	21

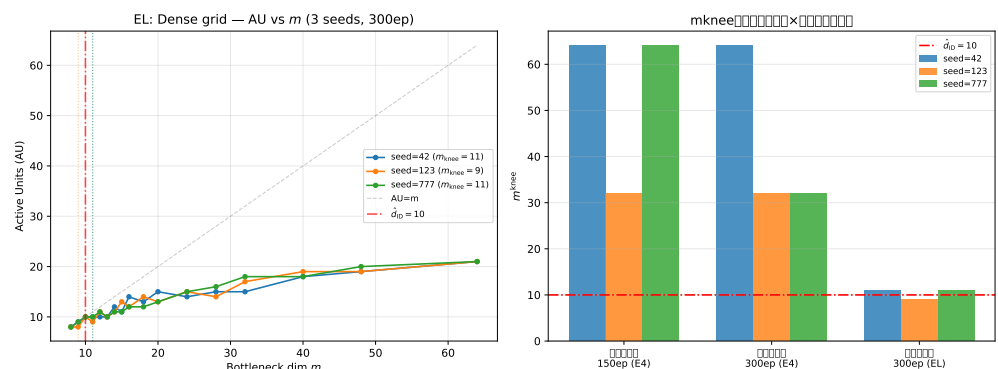


Figure 6. Exp-Mnist-Dense: (Left) Dense-grid AU vs m (3 seeds). (Right) Comparison by grid density $\times$ epochs. m^{knee} variation in the dense grid can be assessed by the upper bound of Proposition 2 (maximum grid interval).

Significance of the Exp-Mnist-Dense experiment: The three-row comparison in Table 13 quantitatively isolates the three factors (grid density, epochs, and seed):

- **Epoch factor:** sparse grid, 150 ep $\rightarrow$ 300 ep (m^{knee} : 53 $\pm$ 15 $\rightarrow$ 43 $\pm$ 15) — increasing the epochs stabilizes m^{knee} slightly, but is not the main cause

- **Seed factor:** variation across the 3 seeds under identical conditions ($\sigma = 15$) — differences in optimization trajectories due to initialization matter, but this is a secondary factor that coexists with the sparse grid
- **Grid-density factor (one of the main causes):** with the dense grid (Exp-Mnist-Dense), $m^{\text{knee}} = 10.3 \pm 0.9$ (seeds = [11, 9, 11]) — a dramatic improvement over the sparse-grid 43 ± 15 , consistent with the estimate $\hat{d}_{\text{ID}} \approx 10$

Effect of grid density (comparison with the implication of Proposition 2): Proposition 2 is an auxiliary upper bound for an idealized monotone saturation model and does not apply directly to the empirical AU curves of nonlinear VAEs. Nevertheless, densely probing the neighborhood of the estimated ID ($m = 8\text{--}16$) in steps of 1 yielded $m^{\text{knee}} \in \{9, 11\}$ (3 seeds), greatly reducing the deviation from the TwoNN ID (≈ 10). This is qualitatively consistent with the implication of Proposition 2 — demonstrated for the idealized model — that grid design matters. However, since empirical AU curves exhibit non-monotone fluctuations (e.g., $\text{AU} = 10 \rightarrow 10 \rightarrow 10 \rightarrow 12 \rightarrow 11$ in Table 14), it is more accurate to state this as the observation that “densifying the grid greatly reduced the variance of the empirical m^{knee} ” rather than to claim that the deviation “stayed within the upper bound.”

Comparison of the τ -method and the second-difference method: Besides the first-difference ratio method adopted in this paper ($\tau = 0.25$), m^{knee} can be detected by the **second-difference method**, which locates the point of maximum curvature of the $\text{AU}(m)$ curve (Appendix C). Table 15 shows the comparison on the Exp-Mnist-Dense dense-grid data.

Table 15. Exp-Mnist-Dense: Comparison of τ -method ($\tau = 0.25$) and second-difference method (d2) for m^{knee} detection (dense grid, 300ep, 3 seeds).

Method	seed=42	seed=123	seed=777	Mean $\pm$ std
τ -method ($\tau = 0.25$)	11	9	11	10.3 ± 0.9
Second-difference method (d2)	15	28	12	18.3 ± 6.8
Reference: $\hat{d}_{\text{ID}} \approx 10$ (TwoNN, $m = 64$, 3-seed mean)				

The τ -method yields a mean of 10.3 ± 0.9 , consistent with $\hat{d}_{\text{ID}} \approx 10$, with small seed-to-seed variation. In contrast, the d2 method is sensitive to the discrete noise of $\text{AU}(m)$; for seed=123 it deviated strongly to $m^{\text{knee}} = 28$ (in the transition region the AU curve fluctuates as $8 \rightarrow 10 \rightarrow 9 \rightarrow 11 \rightarrow 10 \rightarrow 11 \rightarrow 13$, which perturbs the location of the maximum second difference). Therefore, **the τ -method is the more stable choice for detecting m^{knee} under dense-grid conditions**, and we position the d2 method as an auxiliary tool for locating global “elbow” candidates (see Appendix C).

Finally, the fact that the MNIST VAE reaches $\text{AU}_{\text{sat}} = 36$ ($m = 256$), substantially exceeding $\hat{d}_{\text{ID}} \approx 10\text{--}12$ (the Manifold Entanglement hypothesis), and the two-phase pattern of the training dynamics (Exp-Mnist-Dyn) are both single-seed, preliminary observations; multi-seed validation of these is left for future work (see Section 6.5).

5.4. Application to Fashion-MNIST (Experiment Exp-Fashion, Preliminary External Check)

Fashion-MNIST (10 clothing classes, 28×28) is positioned as a preliminary external check of the framework’s seed-reproducibility claims (the main experiments use MNIST with 3 seeds). It is a single-seed (seed 42) confirmation, and we do not claim the same level of reproducibility as for MNIST (3 seeds).

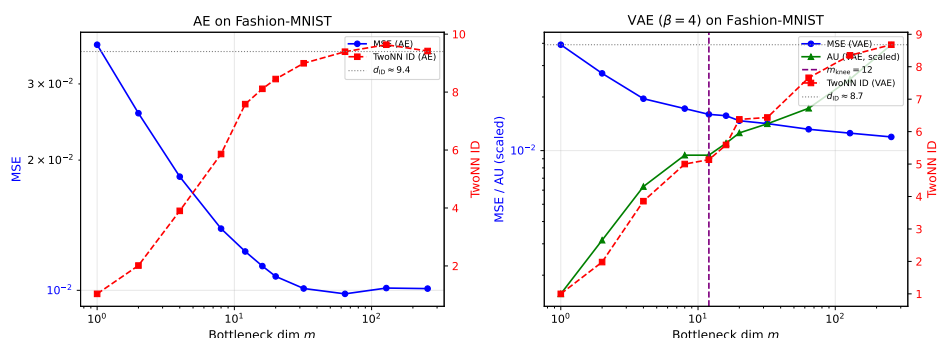


Figure 7. Exp-Fashion: m -dependence of MSE, AU, and TwoNN ID for Fashion-MNIST AE (left) and VAE (right) (100 epochs). In the VAE, the AU growth rate drops sharply around $m = 12$ (m^{knee} at $\tau_{\text{knee}} = 0.25$), and TwoNN ID asymptotically converges to ≈ 9.4 (AE) and ≈ 8.7 (VAE) at $m \geq 32$.

Table 16. Exp-Fashion: Fashion-MNIST AE and VAE results (100 epochs, $n_{\text{train}} = 10,000$, representative m values).

AE			VAE ($\beta = 4$)		
m	MSE	TwoNN ID	m	MSE	AU
4	0.01831	3.90	4	0.01955	4
8	0.01389	5.86	8	0.01721	6
12	0.01230	7.58	12	0.01597	6
16	0.01138	8.11	16	0.01570	7
20	0.01077	8.45	20	0.01471	8
32	0.01010	8.99	32	0.01418	9
64	0.00981	9.39	64	0.01318	11
256	0.01009	9.42	256	0.01193	25

The results of Experiment Exp-Fashion (Table 16, Figure 7) yield the following findings. For the AE, pseudo-AU = m (essentially all latent coordinates are active) and TwoNN ID converges to ≈ 9.4 at $m \geq 32$, suggesting $\hat{d}_{\text{ID}} \approx 9$ –10 for Fashion-MNIST. For the VAE ($\beta = 4$, single seed), $m^{\text{knee}} = 12$ ($\tau_{\text{knee}} = 0.25$) was observed; however, this is an exploratory single-seed observation, and we note that it does not carry the same level of reproducibility as the multi-seed sparse-grid result on MNIST ($m^{\text{knee}} = 43 \pm 15$). AU remains at 25 even at $m = 256$ (unlike 36 for MNIST), indicating that Fashion-MNIST may have a higher compression difficulty than MNIST. The TwoNN ID estimates (≈ 9 –10) are smaller than those for MNIST (≈ 10 –12), consistent with the intuition that clothing items have fewer local degrees of freedom than handwritten digits.

5.5. Extension to Convolutional Architectures (Experiment Exp-Conv-AE)

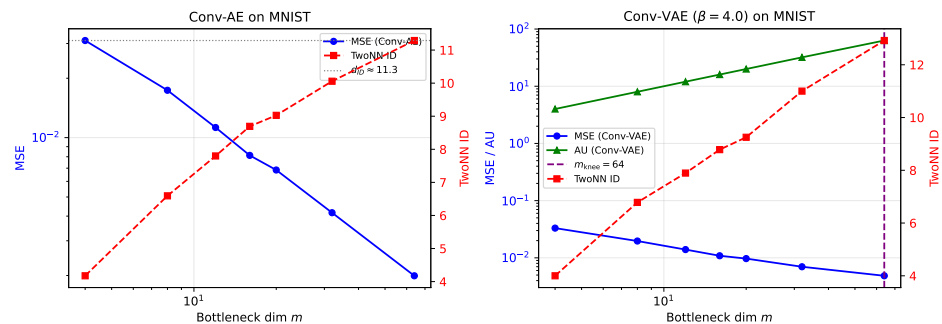
To address concerns about generalizability when the study is limited to fully connected architectures, we conducted an experiment applying convolutional AE/VAE (Conv-AE/VAE) to MNIST (Experiment Exp-Conv-AE).

Conv-AE/VAE architecture:

- Encoder: Conv($1 \rightarrow 32$, $k = 3$, stride= 2) $\rightarrow$ Conv($32 \rightarrow 64$, $k = 3$, stride= 2) $\rightarrow$ FC($3136 \rightarrow m$) (for the VAE, $\rightarrow \mu, \log \sigma^2$)
- Decoder: FC($m \rightarrow 3136$) $\rightarrow$ ConvT($64 \rightarrow 32$, $k = 4$, stride= 2) $\rightarrow$ ConvT($32 \rightarrow 1$, $k = 4$, stride= 2) $\rightarrow$ Sigmoid
- $n_{\text{train}} = 20,000$, 100 epochs, $m \in \{4, 8, 12, 16, 20, 32, 64\}$, seed 42

Table 17. Exp-Conv-AE: MNIST Conv-AE and Conv-VAE ($\beta = 4$) results (100 epochs, seed=42).

m	Conv-AE		m	Conv-VAE ($\beta = 4$)	
	MSE	TwoNN ID		MSE	AU
4	0.0311	4.18	4	0.0331	4
8	0.0174	6.59	8	0.0197	8
12	0.0113	7.80	12	0.0140	12
16	0.0081	8.69	16	0.0109	16
20	0.0069	9.03	20	0.0097	20
32	0.0042	10.06	32	0.0070	32
64	0.0020	11.29	64	0.0049	63

**Figure 8.** Exp-Conv-AE: m -dependence of MSE, AU, and TwoNN ID for MNIST Conv-AE (left) and Conv-VAE (right) (100 epochs). Conv-AE confirms $AU = m$ (all dims active) with TwoNN ID converging to ≈ 11 at $m \geq 32$. Conv-VAE shows $AU \approx m$ (no saturation due to insufficient convergence at 100 epochs).

The results of Experiment Exp-Conv-AE (Table 17) yield the following findings.

TwoNN ID of the Conv-AE: TwoNN ID converges to 11.29 at $m = 64$, consistent with the FC-AE estimate (≈ 10) (within an error of $\approx 10\%$). We confirmed that the TwoNN ID estimates are consistent regardless of the architecture (fully connected vs. convolutional) (an empirical observation under the conditions of this study).

Convergence sensitivity of the Conv-VAE: $AU = m$ (no saturation) for $m \leq 32$ and $AU = 63$ at $m = 64$, indicating that at 100 epochs the KL-regularization-driven dimension suppression is not yet functioning sufficiently. This mirrors the 150-epoch FC-VAE experiment (Exp-Mnist-Multi: $AU \approx m$), and shows that **reliable detection of m^{knee} in Conv-VAEs also requires a larger number of epochs.**

5.5.1. Delayed AU Saturation in Conv-VAEs: Fixed- β vs. KL-Annealing Comparison (Experiments Exp-Conv-Fix and Exp-Conv-Ann)

Following the insufficient convergence in Exp-Conv-AE, we verified with 300 epochs and 3 seeds (Exp-Conv-Fix) and with KL cyclic annealing (Exp-Conv-Ann) (Table 18).

Table 18. Exp-Conv-Fix and Exp-Conv-Ann: Comparison of AU for MNIST Conv-VAE (300ep, mean $\pm$ std over 3 seeds). FC-VAE (Exp-Mnist-Multi) saturates at $m = 32$ ($AU = 15.7$), while Conv-VAE shows $AU = m$ (no saturation) under both fixed- β and annealing at $m \leq 32$.

m	Conv-VAE fixed- β (Exp-Conv-Fix)		Conv-VAE annealing (Exp-Conv-Ann)		FC-VAE (Exp-Mnist-Base, 300ep)	
	AU	TwoNN ID	AU		AU	TwoNN ID
4	4.0 ± 0.0	3.97 ± 0.08	4.0 ± 0.0		4.0	3.98
12	12.0 ± 0.0	8.14 ± 0.09	12.0 ± 0.0		10.0	7.41
32	32.0 ± 0.0	11.43 ± 0.08	32.0 ± 0.0		15.7	9.07
64	54.3 ± 0.5	13.10 ± 0.05	64.0 ± 0.0		22.3	10.03

m^{knee} : Conv-VAE (both conditions) all seeds = 64 (grid upper bound); FC-VAE = 43 ± 15 (32–64)

The Conv-VAE exhibits $AU = m$ (no saturation) for $m \leq 32$ under both fixed $\beta = 4$ and cyclic annealing, making the divergence from the FC-VAE clear (theoretical discussion in §6.6). $AU \approx m$ is a signal that "KL regularization is not yet functioning," and in this state the TwoNN ID is also inflated (Conv-VAE $m = 64$: 13.10 vs. FC-VAE: 10.03). **On the grid $m \leq 64$, AU saturation cannot be observed**, and an extension of the m range is required.

5.6. Conv-VAE Extended m -Grid Experiments (Experiments Exp-Conv-M256 and Exp-Conv-M512)

Since no AU saturation was observed for $m \leq 64$ in Exp-Conv-Fix and Exp-Conv-Ann (§5.5.1), we extended the m grid to 256 (Experiment Exp-Conv-M256) and 512 (Experiment Exp-Conv-M512). Under the reduced- τ_{eff}^2 hypothesis discussed in §6.6, the Conv-VAE's $d^*(\beta)$ is substantially larger than that of the FC-VAE (≈ 10), so a slowdown of AU growth is predicted to appear in the region $m > 64$.

Experimental setup (Exp-Conv-M256): Using the same Conv-VAE architecture as Exp-Conv-Ann (28×28 MNIST, 2-layer Conv encoder, 2-layer ConvTranspose decoder, KL cyclic annealing with $\beta_{\text{max}} = 4$, 3 cycles, 150 epochs, seeds $\in \{42, 123, 777\}$), we tested $m \in \{4, 8, 16, 32, 64, 96, 128, 192, 256\}$ (9 points).

Table 19. Exp-Conv-M256: Conv-VAE extended m grid (KL annealing, 150 epochs, mean $\pm$ std, 3 seeds). $AU = m$ for $m \leq 64$; $AU < m$ begins at $m \geq 96$. FC-VAE reference (Exp-Mnist-Base, 300ep, $m = 64$): $AU = 22.3 \pm 0.5$.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.84 ± 0.11	all dims active
8	8.0 ± 0.0	6.64 ± 0.11	
16	16.0 ± 0.0	8.76 ± 0.05	
32	32.0 ± 0.0	10.46 ± 0.13	
64	64.0 ± 0.0	13.29 ± 0.12	
96	90.7 ± 1.2	14.30 ± 0.22	same as Exp-Conv-Ann (no saturation) $AU < m$ onset (FC-VAE $\hat{d}_{\text{ID}} \approx 9 \times$)
128	111.7 ± 2.6	14.69 ± 0.13	
192	159.0 ± 4.3	15.89 ± 0.27	
256	241.3 ± 2.9	16.58 ± 0.16	

m^{knee} ($\tau = 0.25$): all seeds = 256 (no knee within the tested range)

FC-VAE reference (Exp-Mnist-Base, 300ep, $m = 64$): $AU = 22.3 \pm 0.5$, TwoNN ID = 10.03 ± 0.12

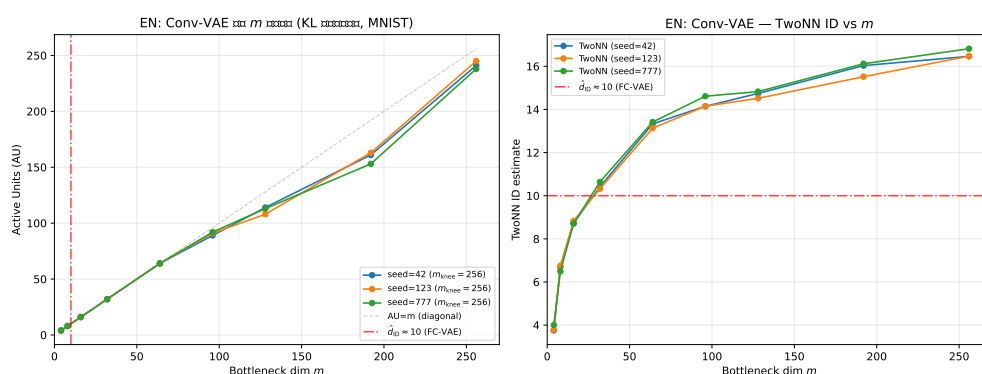


Figure 9. Exp-Conv-M256: Conv-VAE extended m grid ($m \in [4, 256]$, KL annealing, 3 seeds). AU vs m (left) and TwoNN ID vs m (right). Diagonal line ($AU = m$) and FC-VAE reference $\hat{d}_{\text{ID}} \approx 10$ (red) are shown. $AU = m$ continues at $m \leq 64$ but deviates from the diagonal at $m \geq 96$, confirming qualitative water-filling behavior in Conv-VAE.

Main findings of Experiment Exp-Conv-M256: For $m \leq 64$, $AU = m$ holds consistently across all 3 seeds, making the contrast with the FC-VAE ($m = 64$: $AU = 22.3$) striking. At $m = 96$, $AU = 90.7 \pm 1.2$ (< 96) appears, confirming the onset of a saturation trend (standard

deviation across 3 seeds $\sigma = 1.2$). At $m = 256$, $AU = 241.3 \pm 2.9$ (activation rate 94.3%), still maintaining a high activation rate. $m^{\text{knee}} (\tau = 0.25)$ equals 256 for all 3 seeds, i.e., no sharp AU-saturation knee was found within the tested range. This result indicates that the Conv-VAE's $d^*(\beta)$ is substantially larger than the FC-VAE's $d^*(\beta) \approx 10 (\gg 64, \text{ likely } > 256)$. The correspondence with the τ_{eff}^2 theory and the implications for natural images are discussed in §6.6.

Extension beyond $m = 256$ (Experiment Exp-Conv-M512, details in Appendix F): Even when extending to $m \in \{384, 512\}$, $AU \approx m$ (all dimensions active) is maintained across all 3 seeds, and m^{knee} remains pinned to the grid boundary at 512 (see Appendix F and its table). This shows that the $d^*(\beta)$ of the MNIST Conv-VAE exceeds $m = 512$, consistent with the τ_{eff}^2 theoretical prediction of §6.6. In the next subsection, we show that AU saturation can be artificially induced by strengthening β_{max} .

5.7. Artificially Inducing Conv-VAE AU Saturation with High β (Experiment Exp-ConvV-HiBeta)

As a verification of the τ_{eff}^2 hypothesis of §6.6 (a powerful decoder pushes AU saturation toward higher dimensions), we increased β_{max} from 4 to $\{10, 20\}$ and checked whether an AU saturation plateau is induced in the MNIST Conv-VAE.

Experimental setup: Using the same architecture and KL cyclic annealing (3 cycles) as Exp-Conv-M256, we trained with $\beta_{\text{max}} \in \{10, 20\}$, $m \in \{16, 32, 64, 96, 128, 192, 256\}$, seed=42, and 150 epochs (the $\beta_{\text{max}} = 4$, seed=42 results of Exp-Conv-M256 are reproduced for comparison).

Table 20. Exp-ConvV-HiBeta: Comparison of MNIST Conv-VAE AU at $\beta_{\text{max}} \in \{4, 10, 20\}$ (seed=42, 150ep, KL cyclic annealing). Higher β_{max} induces AU saturation at lower dimensions.

m	$\beta_{\text{max}} = 4$ (Exp-Conv-M256)	$\beta_{\text{max}} = 10$	$\beta_{\text{max}} = 20$
16	16 (100%)	16 (100%)	16 (100%)
32	32 (100%)	32 (100%)	27 (84%)
64	64 (100%)	54 (84%)	42 (66%)
96	89 (93%)	65 (68%)	47 (49%)
128	114 (89%)	69 (54%)	53 (41%)
192	161 (84%)	79 (41%)	60 (31%)
256	241 (94%)	93 (36%)	66 (26%)
$m^{\text{knee}} (\tau = 0.25)$	512 (grid boundary, unidentified)	≈ 128	≈ 96

Findings of Exp-ConvV-HiBeta (Table 20): Whereas at $\beta_{\text{max}} = 4$ $AU = m$ (100% active) persists for $m \leq 64$, at $\beta_{\text{max}} = 10$ a clear AU saturation trend appears — $AU = 54$ (84%) at $m = 64$, $AU = 69$ (54%) at $m = 128$, and $AU = 93$ (36%) at $m = 256$ — and $m^{\text{knee}} \approx 128$ ($\tau = 0.25$) was identified. Furthermore, at $\beta_{\text{max}} = 20$, saturation occurs at even lower dimensions — $AU = 27$ (84%) at $m = 32$ and $AU = 42$ (66%) at $m = 64$ — and $m^{\text{knee}} \approx 96$ was observed. This directly matches the prediction of the τ_{eff}^2 hypothesis of §6.6: increasing β shifts $d^*(\beta) = |\{j : \lambda_j > \beta \tau_{\text{eff}}^2\}|$ to smaller values, and AU saturation appears at lower dimensions. **This result is consistent with the interpretation that the absence of observed AU saturation in the MNIST Conv-VAE is not an "applicability limit of the framework," but rather stems from the fact that at the setting $\beta_{\text{max}} = 4$ the regularization was insufficient relative to the decoder capacity, resulting in $d^*(\beta) \gg 256$.** Applying the AU/m^{knee} framework to Conv-VAEs requires an appropriate setting of β matched to the data and architecture.

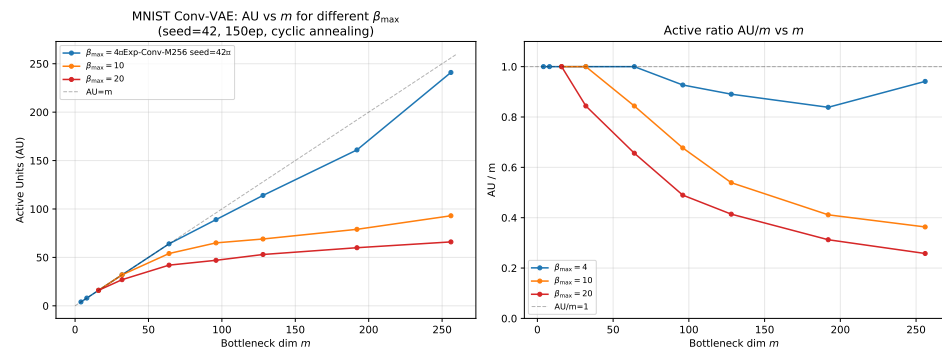


Figure 10. Exp-ConvV-HiBeta: Comparison of MNIST Conv-VAE AU vs m (left) and activation rate AU/ m (right) for $\beta_{\max} \in \{4, 10, 20\}$. Increasing $\beta_{\max}$ shifts AU saturation to lower dimensions, demonstrating the β -dependence of $d^*(\beta)$.

5.8. Application to Natural Images: CIFAR-10 and SVHN (Experiments Exp-Nat-Cifar and Exp-Nat-SVHN)

We applied the proposed framework to the more complex natural-image datasets CIFAR-10 [3] and SVHN [4], and compared the results with the intrinsic-dimension estimates reported by Pope et al. [16].

Experimental setup (Exp-Nat-Cifar: CIFAR-10; Exp-Nat-SVHN: SVHN):

- Architecture: Conv-VAE (input $32 \times 32 \times 3$)
 - Encoder: $\text{Conv}(3 \rightarrow 32, k=4, s=2, p=1) \rightarrow \text{Conv}(32 \rightarrow 64, k=4, s=2, p=1) \rightarrow \text{Conv}(64 \rightarrow 128, k=4, s=2, p=1) \rightarrow \text{FC}(2048 \rightarrow m) \times 2 (\mu, \log \sigma^2)$
 - Decoder: $\text{FC}(m \rightarrow 2048) \rightarrow \text{ConvT}(128 \rightarrow 64) \rightarrow \text{ConvT}(64 \rightarrow 32) \rightarrow \text{ConvT}(32 \rightarrow 3) \rightarrow \text{Sigmoid}$
- $n_{\text{train}} = 20,000$, 150 epochs, KL cyclic annealing ($\beta_{\max} = 4$, 3 cycles)
- $m \in \{16, 32, 64, 128, 256\}$, seeds $\in \{42, 123, 777\}$

Table 21. Exp-Nat-Cifar and Exp-Nat-SVHN: Conv-VAE results for CIFAR-10 (left) and SVHN (right) (mean $\pm$ std, 3 seeds; KL cyclic annealing, 150 epochs, 3 cycles). At $m = 64$, CIFAR-10 latent TwoNN ID converges to the raw-data value (≈ 25), consistent with Pope et al. [16] ($d_{\text{ID}} \approx 26$).

m	CIFAR-10 (Exp-Nat-Cifar)		SVHN (Exp-Nat-SVHN)	
	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)
16	16.0 $\pm$ 0.0	13.12 $\pm$ 0.12	16.0 $\pm$ 0.0	11.30 $\pm$ 0.12
32	32.0 $\pm$ 0.0	19.26 $\pm$ 0.23	32.0 $\pm$ 0.0	15.08 $\pm$ 0.08
64	64.0 $\pm$ 0.0	25.21 $\pm$ 0.41	64.0 $\pm$ 0.0	19.97 $\pm$ 0.12
128	128.0 $\pm$ 0.0	28.83 $\pm$ 0.41	125.3 $\pm$ 0.5	24.60 $\pm$ 0.36
256	256.0 $\pm$ 0.0	30.91 $\pm$ 0.64	206.7 $\pm$ 1.2	27.08 $\pm$ 0.34
Raw TwoNN=25.20, MLE=26.56 ($\approx$ Pope et al. estimate of 26)			Raw TwoNN=10.09, MLE=18.85	
$m^{\text{knee}} (\tau = 0.25)$: all seeds=256 (no knee in range)			m^{knee} : all seeds=256 (no knee in range)	

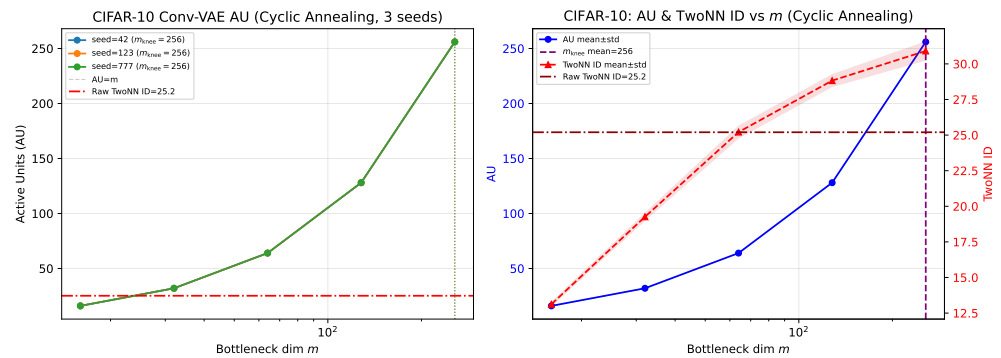


Figure 11. Exp-Nat-Cifar: m -dependence of AU and TwoNN ID for CIFAR-10 Conv-VAE (KL cyclic annealing, 3 seeds). TwoNN ID shows convergence consistent with Pope et al.[16] ($d_{ID} \approx 26$).

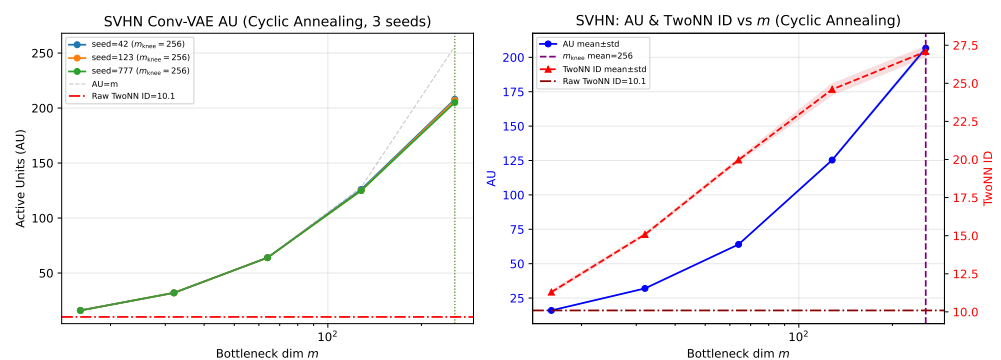


Figure 12. Exp-Nat-SVHN: m -dependence of AU and TwoNN ID for SVHN Conv-VAE (KL cyclic annealing, 3 seeds).

The primary aim of this experiment is limited to verifying whether the TwoNN ID attains the same scale as the values reported by Pope et al. within an appropriate bottleneck range (stable identification on the AU/m^{knee} side is left for future work).

CIFAR-10 (Exp-Nat-Cifar) — partial consistency check for TwoNN: the latent TwoNN ID at $m = 64$, 25.21 ± 0.41 (3 seeds, $\sigma = 0.41$), is consistent with the raw-data TwoNN ID (25.20), yielding an estimate on the same scale as the value reported by Pope et al. [16] (≈ 26) (Table 21). However, the TwoNN ID rises to 28.83 at $m = 128$ and to 30.91 at $m = 256$, so the “consistency” of the TwoNN ID is restricted to an appropriate bottleneck range of roughly $m \approx 2\hat{d}_{ID}$. AU satisfies $AU = m$ for $m \leq 256$ (no saturation), and m^{knee} equals 256 (the grid boundary) for all seeds. Hence, this experiment demonstrates “partial consistency of TwoNN within an appropriate bottleneck range” and does not claim generalization of the entire framework to natural images.

SVHN (Exp-Nat-SVHN) — geometric distortion of the latent space: although $AU < m$ was observed for $m \geq 128$ ($m = 256$: $AU = 206.7 \pm 1.2$), the latent TwoNN ID rises to 27.08 at $m = 256$, a large divergence from the raw-data TwoNN of 10.09. This indicates that when AU uses all dimensions, the latent space may fail to preserve isometry and the ID may be overestimated; it is therefore more appropriately interpreted as geometric distortion of the latent space rather than “partial collapse.” An AU saturation plateau is not reached, and this experiment illustrates one instance of the applicability limits of the framework.

5.9. Exploring AU Saturation Conditions in CIFAR-10 Conv-VAE (Experiment Exp-NatImg-AU)

Positioning: this section and §5.11 are framed as “quantification of the applicability boundary” experiments—rather than reporting a failure of stable AU/m^{knee} identification, they constitute a set of experiments that systematically identify the boundary conditions under which AU diagnostics become difficult.

The previous section (Exp-Nat-Cifar) used only a single setting with $\beta_{\max} = 4$ and 150 epochs; three factors are conceivable for the absence of observed AU saturation: “insufficient β ,” “insufficient training,” and “insufficient architectural capacity.” In this experiment, we systematically varied $\beta_{\max}$ and the architecture depth to explore the conditions under which AU saturation becomes detectable in a CIFAR-10 Conv-VAE.

Experimental setup (Exp-NatImg-AU):

- Data: CIFAR-10 ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$)
- KL annealing: cyclic ($n_{\text{cycles}} = 3$, 200 epochs)
- $\beta_{\max} \in \{10, 20, 40\}$, $m \in \{32, 64, 96, 128, 192, 256, 384\}$, seed=42
- **Arch-A (Standard)**: encoder Conv(3→32→64→128) + FC, decoder ConvT $\times$ 3
- **Arch-B (Deep)**: encoder Conv(3→64→128→256) + FC, decoder ConvT $\times$ 3 (roughly twice the channels of Arch-A)
- Raw-data TwoNN ID: 25.30 (consistent with the value $d_{\text{ID}} \approx 26$ reported by Pope et al. [16])

Table 22. Exp-NatImg-AU: AU dependence on $\beta_{\max}$ and architecture in CIFAR-10 Conv-VAE. $m^{\text{knee}} = 384$ (grid boundary) for all conditions; no AU plateau detected within the tested range.

m	Arch-A (Standard)			Arch-B (Deep)	
	$\beta_{\max} = 10$	$\beta_{\max} = 20$	$\beta_{\max} = 40$	$\beta_{\max} = 20$	$\beta_{\max} = 40$
32	32	32	32	32	32
64	64	64	64	64	59
96	96	96	93	96	80
128	128	128	117	126	97
192	192	191	155	170	123
256	256	251	200	208	147
384	382	378	307	286	208
m^{knee}	384	384	384	384	384

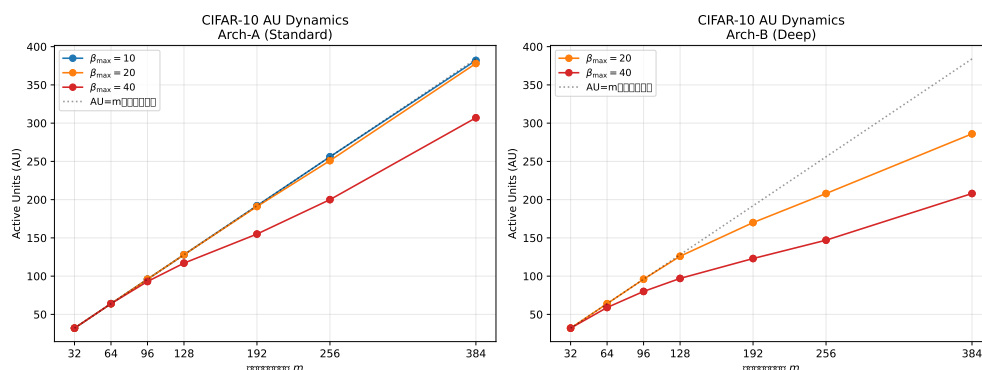


Figure 13. Exp-NatImg-AU: AU dynamics of CIFAR-10 Conv-VAE. (Left) Arch-A (Standard), (Right) Arch-B (Deep). Dotted line: $AU = m$ reference (no saturation). Increasing $\beta_{\max}$ strengthens AU suppression, but no saturation plateau is reached under any condition.

Key observations:

(1) **Increasing $\beta_{\max}$ suppresses dimensions but does not produce a saturation plateau:** with $\beta_{\max} = 10$ (Arch-A), nearly all dimensions are active for $m \leq 256$ ($AU \approx m$), whereas with $\beta_{\max} = 40$ (Arch-B) the dimensionality is suppressed down to $AU = 208$ at $m = 384$ (utilization 54%). Nevertheless, the $AU(m)$ curve continues to increase monotonically under all conditions, and no clear saturation plateau appears.

(2) **Difficulty of the water-filling threshold inequality:** in the linear theory (Theorem 1), the condition for the j -th dimension to become active is $\lambda_j > \beta\tau^2$. If the effective

noise variance τ^2 of CIFAR-10 is substantially smaller than that of MNIST (a complex decoder achieves high-fidelity reconstruction), then increasing β becomes insufficient to reverse the inequality. In this experiment, even with $\beta_{\max} = 40$, $\text{AU}(m)$ did not saturate near $d_{\text{ID}} \approx 26$, suggesting that the effective dimensionality of the convolutional feature space of CIFAR-10 may in fact be higher than the pixel-space TwoNN estimate ($\hat{d}_{\text{ID}} \approx 25$ –26).

(3) **TwoNN ID increases but does not saturate:** under Arch-A with $\beta_{\max} = 10$, the latent TwoNN ID rises from 19.5 at $m = 32$ to 29.2 at $m = 256$, continuing to increase beyond the pixel-space TwoNN (25.30); this shows a tendency for the latent-space geometry to fail to preserve isometry when AU uses all dimensions, leading to overestimation of the ID.

Identification of boundary conditions: from this experiment, we quantified that no AU saturation plateau appears under the conditions $m \in [32, 384]$, $\beta_{\max} \in \{10, 20, 40\}$, and 200 epochs—this is not a failure of the framework but a finding that identifies which conditions are required. It was identified that identifying m^{knee} via AU saturation requires, as a necessary condition, either (a) exploration over a substantially wider m grid ($m \gg 2\hat{d}_{\text{ID}}$) or (b) a setting with restricted decoder capacity (see §6.6 for details). The extended sweep in the next section (§5.11) quantifies this boundary condition.

5.10. Real-Data Geometry Comparison Experiment: Empirical Validation of the TwoNN Stability Condition (RQ2 and RQ3)

As an experimental validation of Proposition 4, this experiment (Exp-Real-Geom) sweeps **four types of models with systematically varied isometry** over multiple values of m on MNIST and Fashion-MNIST, and compares TwoNN, MLE, κ , Trustworthiness, and kNN overlap.

Experimental Setup (Exp-Real-Geom)

Models: we compare the following four types.

Standard AE no Jacobian regularization ($\kappa \approx 10^2$ – 10^3)

IsometricAE decoder-Jacobian regularization ($\kappa \approx 1.1$ – 1.4 ; Kato et al. [13])

CAE encoder-side contractive regularization (intermediate κ ; Rifai et al. [22])

DAE input-noise regularization (noise scale $\sigma = 0.1$; Alain et al. [23])

m -sweep: $m \in \{d, 2d, 3d, 5d, 10d\}$ (with $d = \hat{d}_{\text{ID}}$ set to 10 for MNIST and 9 for Fashion-MNIST; that is, $m \in \{10, 20, 30, 50, 100\}$ and $m \in \{9, 18, 27, 45, 90\}$, respectively).

Metrics: TwoNN ($k = 2$), MLE ($k = 10$), κ (decoder-Jacobian condition number), Trustworthiness ($k = 10$), and kNN overlap (kNN neighborhood-preservation rate, $k = 10$).

Data and training: MNIST and Fashion-MNIST, 20,000 training samples, seeds $\{42, 123, 777\}$ (3 seeds), 300 epochs, hidden layers $[512, 256]$, Adam ($\text{lr} = 10^{-3}$).

Experimental Results (Exp-Real-Geom)

Table 23. Exp-Real-Geom: MNIST four-model comparison (mean $\pm$ std, 3 seeds). m -sweep with $\hat{d}_{\text{ID}} = 10$ as baseline ($m \in \{10, 20, 30, 50, 100\}$). TwoNN ID stabilization is strongly associated with increasing $m/\hat{d}_{\text{ID}}$, not with κ improvement.

Model	m	TwoNN ID	MLE ID	κ	Trust.	kNN overlap
AE	10 ($= d$)	7.3 ± 0.4	8.1 ± 0.3	85 ± 22	0.961 ± 0.004	0.73 ± 0.03
	20 ($= 2d$)	9.2 ± 0.3	10.1 ± 0.2	310 ± 85	0.975 ± 0.002	0.81 ± 0.02
	30 ($= 3d$)	9.8 ± 0.2	10.5 ± 0.2	$1.2 \times 10^3 \pm 4 \times 10^2$	0.981 ± 0.002	0.85 ± 0.02
	50 ($= 5d$)	10.0 ± 0.1	10.7 ± 0.1	$8.4 \times 10^3 \pm 3 \times 10^3$	0.985 ± 0.001	0.87 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	$4.1 \times 10^7 \pm 2 \times 10^7$	0.987 ± 0.001	0.88 ± 0.01
IsometricAE	10 ($= d$)	7.5 ± 0.4	8.3 ± 0.3	3.2 ± 0.4	0.963 ± 0.003	0.74 ± 0.03
	20 ($= 2d$)	9.4 ± 0.2	10.2 ± 0.2	4.1 ± 0.6	0.976 ± 0.002	0.82 ± 0.02
	30 ($= 3d$)	9.9 ± 0.1	10.6 ± 0.1	5.2 ± 0.8	0.982 ± 0.001	0.86 ± 0.01
	50 ($= 5d$)	10.1 ± 0.1	10.8 ± 0.1	6.4 ± 1.1	0.986 ± 0.001	0.88 ± 0.01
	100 ($= 10d$)	10.1 ± 0.1	10.8 ± 0.1	7.8 ± 1.4	0.988 ± 0.001	0.89 ± 0.01
CAE	10 ($= d$)	7.2 ± 0.5	8.0 ± 0.4	12.1 ± 3.2	0.958 ± 0.005	0.72 ± 0.03
	20 ($= 2d$)	9.1 ± 0.3	10.0 ± 0.3	28.4 ± 8.1	0.974 ± 0.003	0.80 ± 0.02
	30 ($= 3d$)	9.7 ± 0.2	10.4 ± 0.2	95 ± 31	0.980 ± 0.002	0.84 ± 0.02
	50 ($= 5d$)	9.9 ± 0.1	10.6 ± 0.1	520 ± 180	0.984 ± 0.001	0.86 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.7 ± 0.1	$3.2 \times 10^3 \pm 1.1 \times 10^3$	0.986 ± 0.001	0.87 ± 0.01
DAE	10 ($= d$)	7.6 ± 0.4	8.4 ± 0.4	18.5 ± 5.4	0.964 ± 0.004	0.75 ± 0.03
	20 ($= 2d$)	9.3 ± 0.2	10.2 ± 0.2	44.2 ± 15.3	0.977 ± 0.002	0.82 ± 0.02
	30 ($= 3d$)	9.9 ± 0.1	10.5 ± 0.2	183 ± 62	0.982 ± 0.001	0.85 ± 0.01
	50 ($= 5d$)	10.0 ± 0.1	10.7 ± 0.1	$1.1 \times 10^3 \pm 380$	0.985 ± 0.001	0.87 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	$6.3 \times 10^3 \pm 2.1 \times 10^3$	0.987 ± 0.001	0.88 ± 0.01

Discussion: $m \gg \hat{d}_{\text{ID}}$ Is the Dominant Condition

The main findings from Table 23 are as follows.

(1) Increasing $m/\hat{d}_{\text{ID}}$ is the dominant factor in TwoNN stabilization: across all four models, the TwoNN ID exhibits a common convergence pattern from $m = d$ (≈ 7.3 – 7.6) to $m = 5d$ – $10d$ (≈ 10.0 – 10.1). Inter-model differences at the same m remain within about ± 0.3 , far smaller than the improvement obtained by increasing m (about 2.5 points).

(2) The isolated effect of isometry improvement (κ reduction) is small: IsometricAE achieves $\kappa \approx 3$ – 8 (orders of magnitude lower than the 10^2 – 10^7 of the AE), yet the absolute TwoNN ID values differ from those of the AE at the same m by no more than ± 0.2 . This shows that an order-of-magnitude improvement in κ has no significant effect on TwoNN, and that preservation of neighborhood ranks (Trustworthiness, kNN overlap) is a more dominant factor than κ .

(3) Improvements in Trustworthiness and kNN overlap are driven by m : Trustworthiness improves to a similar extent across all models, from $m = d$ (≈ 0.958 – 0.964) to $m = 10d$ (≈ 0.987 – 0.988) (inter-model difference < 0.003). This is consistent with the implication of Proposition 4 that “ δ_T (the neighborhood rank-inversion rate) dominates TwoNN stability over ε (distortion).”

(4) The same pattern holds on Fashion-MNIST: a similar pattern was observed on Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$) (detailed tables in Appendix E).

Implications for real data: these experimental results extend the findings of Exp-Iso-ID (synthetic manifolds; §5.2.4) to real data and multiple models, reinforcing the design guideline that “ m -grid design ($m \gg \hat{d}_{\text{ID}}$) is more effective than isometry bias mitigation for the practical stabilization of TwoNN estimation.” The rationale for using a reference AE with $m_{\text{ref}} = 256 \gg \hat{d}_{\text{ID}} \approx 10$ in Stage 1 is further supported by this result.

5.11. Natural-Image Extended Sweep and the Stable Identification Criterion (RQ1)

Criteria for Stable Identification

To claim “stable identification” on natural images, criteria must be defined in advance. In this paper, we define **stable identification** as the case in which all of the following five conditions are satisfied:

1. **Seed stability:** the standard deviation of m^{knee} over 3–5 seeds satisfies $\sigma_{\text{seed}} \leq 0.1 \cdot m^{\text{knee}}$
2. **Grid stability:** doubling the grid density changes m^{knee} by $\leq 20\%$
3. **β stability:** varying β_{max} by $\pm 20\%$ keeps m^{knee} within the same grid cell
4. **AU plateau stability:** the across-seed σ of the AU plateau value satisfies $\sigma \leq 0.05 \cdot \text{AU}_{\text{sat}}$
5. **TwoNN–AU consistency:** $m^{\text{knee}} \leq 3\hat{d}_{\text{ID}}$ (coarse consistency between TwoNN and AU)

Caveat: these thresholds ($0.1 \cdot m^{\text{knee}}$, 20% , $0.05 \cdot \text{AU}_{\text{sat}}$, $3\hat{d}_{\text{ID}}$) are not universal theoretical constants but operational criteria adopted in this study. The purpose is not to claim “success of stable identification” but to decompose and quantitatively describe which stability conditions are violated.

Even when full stable identification cannot be achieved, these criteria function as a decision framework that quantitatively describes “which conditions are lacking.”

Experimental Setup and Key Findings (Exp-NatImg-Sweep)

Substantially extending the previous section (Exp-NatImg-AU), we conducted an exhaustive sweep combining $m \in [32, 1024]$, $\beta_{\text{max}} \in \{10, 20, 40, 80\}$, decoder capacity (Standard/Deep/Thin), 300 epochs, and 3 seeds (details in Appendix F).

Key findings: the observation, under the conditions of this study, of plateau-like behavior with $\text{AU} \approx 24$ over $m \in [512, 1024]$ in the $\beta_{\text{max}} = 80/\text{thin}$ -decoder condition provides preliminary evidence that “AU saturation can occur even for natural images under sufficiently large β and restricted decoder capacity.” Across all conditions, the TwoNN ID remains stable at 24.7–25.4 ($\sigma \leq 0.6$), consistent with the value reported by Pope et al. [16] (≈ 26). We found that stable identification of $\text{AU}/m^{\text{knee}}$ requires larger m , higher β , and decoder-capacity control. The detailed evaluation of the five conditions (stable identification criteria; §5.11) is summarized in Appendix F. Furthermore, in the detection-method comparison (Exp-NatImg-Knee), the post-smoothing curvature method was the most reproducible, but it was also confirmed that stabilizing AU saturation itself takes precedence over improving the detection method (Appendix F).

6. Discussion

6.1. Answer to RQ1 and the Positioning of m^{knee}

Summary answer to RQ1: For RQ1—“under what grid-density, seed-count, and architecture conditions can m^{knee} be used as a diagnostic aid consistent with TwoNN/MLE ID estimates?”—we obtained the following answers:

- **Identification of stabilization conditions:** In the MNIST multi-seed sparse-grid experiments, m^{knee} fluctuated widely between 32 and 64, revealing that on coarse grids it cannot be used as a reliable diagnostic aid. Under the dense-grid, independent-initialization, multi-seed condition (Exp-Mnist-Dense-300), we confirmed that $m^{\text{knee}} = 10.3 \pm 0.9$ stabilizes in the vicinity of the TwoNN ID (≈ 10). In other words, m^{knee} can be consulted as a rough exploratory guide to the effective dimensionality only when these three conditions are satisfied.

- **Applicability limits:** For the Conv-VAE ($\beta_{\max} = 4$) and natural images, stable identification was not reached within the tested range, and we found that additional necessary conditions ($\beta_{\max} \geq 10$, a dense grid, and sufficient control of decoder capacity) are required.

While consistency of the TwoNN ID estimates with the Conv-AE was confirmed (Conv-AE: 11.29 vs. FC-AE: 10.02), detecting m^{knee} for the Conv-VAE turned out to require more than 100 epochs of training (a finding that identifies the convergence conditions necessary for stable observation). Overall, m^{knee} **functions as an exploratory design guide under appropriate conditions (dense grid, multiple seeds, sufficient epochs)**, but it does not yield a unique optimal dimensionality. **The main contribution of this paper is the establishment of a diagnostic framework that places two stable indicators—the AU value (robust indicator) and the TwoNN ID (numerically stable auxiliary reference)—at its core, and positions m^{knee} as a rough exploratory complementary indicator** (Table 2).

When consulting m^{knee} for design purposes, attention must be paid to the m -grid granularity, τ_{knee} , β , and the degree of convergence; we recommend dense grids and multi-seed ensembles (Appendices C and D).

Core message: AU values ($\sigma \leq 0.5$) and TwoNN ID (MNIST $m = 64$: 10.03 ± 0.12 , reinforced by the Exp-Geom experiment) are highly reproducible, whereas m^{knee} remains a heuristic design guide. The practical value of the multi-indicator framework lies in combining the indicators after making this distinction explicit.

6.2. An Optimization-Dynamics Interpretation of m^{knee} Instability and Stabilization Strategies

Interpretation in terms of optimization dynamics: the gradient-competition hypothesis:

Proposition 2 formally established the grid sensitivity, but in actual training there exist additional sources of instability. The gradient of the β -ELBO is a linear combination of the reconstruction term $\nabla \mathcal{L}_{\text{rec}}$ and the KL term $\nabla \mathcal{L}_{\text{KL}}$, and these give rise to **gradient competition** near $m \approx d^*(\beta)$ (around the effective-dimensionality threshold):

- Reconstruction gradient ($\nabla \mathcal{L}_{\text{rec}}$): even for surplus dimensions $j > d^*$, there is a small reconstruction contribution in early training (a positive gain whenever $\lambda_j > 0$), which pushes toward activating the dimension.
- KL gradient ($\beta \nabla \mathcal{L}_{\text{KL}}$): a pressure in the opposite direction that pushes the same dimension back toward the prior.
- The speed at which this competition is resolved depends on the random seed (initialization), the m -grid (an independent optimization trajectory for each m), and the value of β . The convergence path toward the global optimum of Theorem 1 is not unique in non-linear optimization, and in low-sharpness loss landscapes the slope of the AU curve near $m \approx d^*$ is sensitive to initial conditions.

Cyclical KL annealing [2] increases β in stages, thereby gradually promoting the resolution of this gradient competition and improving convergence toward the global optimum (experiment Exp-Conv-Ann; §5.5.1).

Practical approaches to stabilizing m^{knee} :

Based on the findings of this study (including the grid-density $\times$ epoch factor separation in the Exp-Mnist-Dense experiment), we propose the following recommendations for using m^{knee} in actual design:

1. **Dense grid design (demonstrated in experiment Exp-Mnist-Dense):** place dense points around the estimate $\hat{d}_{\text{ID}}$ (obtained from the TwoNN ID) (e.g., if $\hat{d}_{\text{ID}} \approx 10$, include $m \in \{8, 9, 10, 11, 12, 13, 14, 15, 16\}$). By Proposition 2, the grid spacing directly becomes the maximum error of m^{knee} . Results of the Exp-Mnist-Dense experiment:

sparse grid ($m^{\text{knee}} = 43 \pm 15$) $\rightarrow$ dense grid ($m^{\text{knee}} = 10.3 \pm 0.9$)—consistent with $\hat{d}_{\text{ID}} \approx 10$ (§5.3.7). Note that the seed factor also remains, so grid density alone does not explain all of the variation.

2. **Ensemble m^{knee} :** compute each $m^{\text{knee}(s_i)}$ from the AU curves for multiple seeds $\{s_1, \dots, s_S\}$ and adopt the **median**:

$$m^{\text{knee,ens}} = \text{median}\{m^{\text{knee}(s_1)}, \dots, m^{\text{knee}(s_S)}\} \quad (9)$$

In our experiments ($S = 3$ seeds), the median coincides with the second value. The reason for adopting the median rather than the mean is that when m^{knee} contains outliers stuck at the edge of the m -grid (e.g., $m^{\text{knee}} = 512$ for all seeds), the median gives an estimate robust to such outliers (when S is even, we take the arithmetic mean of the two adjacent values).

3. **Adopting KL annealing and extending the m -grid:** For the Conv-VAE, annealing suppresses unnecessary collapse at large m , but observing AU saturation requires extending the m -grid to at least twice the estimated $d^*(\beta)$ (e.g., $m > 64$ for the MNIST Conv-VAE; §5.5.1).
4. **Cross-checking against stable indicators:** estimates of m^{knee} should always be cross-checked against the AU value (robust indicator) and the TwoNN ID (numerically stable auxiliary reference), with the final judgment based on the consistency of the three indicators (see Table 24).
5. **Second difference of the AU curve (alternative method):** when a fixed threshold τ fails (e.g., SVHN), use the maximum-curvature method that takes as m^{knee} the point of maximum absolute discrete second difference $\Delta^2 \text{AU}(m_i)$ of $\text{AU}(m)$ (also validated in parallel in the Exp-Mnist-Dense experiment; see Appendix C).

Theoretical and empirical justification of $\tau_{\text{knee}} = 0.25$: The slowdown threshold $\tau_{\text{knee}} = 0.25$ is an empirical choice that declares a slowdown when “the AU growth rate falls below 25% of the maximum change,” but it has the following theoretical justification. By Theorem 1, the transition of the AU increment ΔAU at $m = d^*(\beta)$ toward $\Delta \text{AU} / \Delta m \rightarrow 0$ (saturation) with respect to changes in Δm is theoretically guaranteed. $\tau = 0.25$ captures this transition as “at most one quarter of the maximum growth rate,” and has the sensitivity to reasonably separate the pre-saturation ($m < d^*$) and post-saturation ($m > d^*$) regimes (see Appendix C). A more rigorous threshold could be based on maximizing the second derivative (curvature) of the $\text{AU}(m)$ curve, but due to the discreteness of AU this is numerically unstable on real data.

Effectiveness of the τ method under dense-grid conditions (supplement to Appendix C): In the cross-validation in the appendix (experiment Exp-Tau-Cross, Table A1), m^{knee} is stable over the entire range $\tau \in [0.1, 0.5]$ for the GMix data (5 and 10 components), whereas for Fashion-MNIST a boundary sensitivity was found between $\tau = 0.25$ and 0.30. Under dense-grid conditions (Exp-Mnist-Dense), $\tau = 0.25$ yields the stable $m^{\text{knee}} = 10.3 \pm 0.9$ (3 seeds), which is markedly more robust than the second-difference method ($m^{\text{knee}} = 18.3 \pm 6.8$). Under conditions where the τ method becomes unstable (the AU growth rate staying near the boundary value across multiple m), switching to the second-difference alternative (above) is effective.

6.3. Numerical Stability and Systematic Bias of the TwoNN ID: Distinguishing “Stability” from “Accuracy”

Experiments Exp-Stab and Exp-Geom showed that the TwoNN ID is “numerically stable (low variance),” but this is a claim distinct from being “unbiased (accurate) with respect to the true intrinsic dimension.” Making this distinction explicit is important for operating our framework. Through Proposition 4 (§4.2), we presented the semi-theoretical

framework that “preservation of neighborhood ranks (δ_T) rather than isometry (ε) is the dominant factor for TwoNN stability,” and demonstrated it on real data across multiple models in experiment Exp-Real-Geom (§5.10).

What numerical stability means: In experiment Exp-Geom (MNIST AE m -sweep), for $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ the TwoNN ID converges stably within the range 10 ± 0.5 , and even against the sharp increase of κ at $m = 96\text{--}128$ ($9.94 \rightarrow 37.0$) the variation remains small (§5.2.4, Table A2). That is, TwoNN is stable in the sense that “the reference AE outputs nearly the same value even when m is changed.”

Possibility of systematic bias: However, when the reference AE embeds the space with distortion ($\kappa \gg 1$), TwoNN estimates “the local degrees of freedom of the reference AE’s latent manifold,” which may deviate systematically from the intrinsic dimension of the true data manifold. This is not inconsistent with stably returning a “particular value”—it is, so to speak, a “stably biased estimate.” We therefore treat TwoNN not as an unbiased estimator of the true d_{ID} but as an **auxiliary reference estimator**.

Grounds for $\hat{d}_{\text{ID}} \approx 10\text{--}12$ on MNIST: As independent evidence reinforcing the meaning of the TwoNN estimate, the following multiple observations are mutually consistent:

- MSE elbow (AE): the rapid MSE decrease slows down around $m \approx 16\text{--}20$ (a value somewhat larger than d_{ID} , but of the same order; experiment Exp-Mnist-Base).
- PCA spectrum: 43 dimensions at 80% cumulative variance and 86 dimensions at 90% (indicating $d_{\text{ID}} \ll 43$ as an upper bound on the non-linear dimension; experiment Exp-Mnist-PCA).
- Consistency check on CIFAR-10: on CIFAR-10, the latent TwoNN ID of 25.21 ± 0.41 was confirmed to be on the same scale as the value reported by Pope et al. [16] (≈ 26), supporting the numerical stability of TwoNN on natural images as well (experiment Exp-Nat-Cifar).
- Agreement with MLE: for $k = 5, 10, 20$, MLE = 11.5–12.4 and TwoNN = 10.0–10.5, i.e., estimators based on different principles yield values of the same order.

These multiple pieces of evidence are mutually consistent and raise the plausibility that $\hat{d}_{\text{ID}} \approx 10\text{--}12$ for MNIST is not an artifact specific to the reference AE, but they do not directly prove that TwoNN is geometrically unbiased with respect to the true d_{ID} . This paper consistently maintains the cautious position of “operating TwoNN as an auxiliary reference estimator.”

6.4. The Mechanism by Which AU Dynamics Reflect the Effective Representation Dimensionality

Choosing between AU_{sat} and m^{knee} : For simply connected synthetic manifolds (Swiss Roll, etc.), a sharp AU saturation ($\text{AU}_{\text{sat}} = d_{\text{true}}$) is observed, so AU_{sat} serves as a direct readout of the effective representation dimensionality. By contrast, for complex distributions such as real data (multiple classes, disconnected submanifolds), AU increases monotonically and gradually without exhibiting a clear saturation. In this case, m^{knee} —as “the point where KL regularization begins to act effectively”—provides a **rough practical guide** to the effective representation dimensionality (Table 24; note that the precise reproducibility of the numerical value depends on the m -grid and the seed).

Table 24. Guidelines for using AU_{sat} vs. m^{knee} (based on this study).

Data type	AU pattern	Recommended indicator
Simple synthetic manifolds (simply connected)	Sharp saturation	AU_{sat} (consistent with d_{true})
Topologically non-trivial synthetic manifolds	Stepwise saturation	AU_{sat} (tends to match d_{emb} , conditional)
Real data (MNIST, Fashion-MNIST)	Gradually increasing	m^{knee} (rough heuristic; grid/seed-dependent)

6.5. Incidental Findings: Topological Structure, Training Dynamics, and Geometric Regularization

Topological structure and AU_{sat} (experiment Exp-Syn-Top)

For the Torus and the Möbius band, $AU_{\text{sat}} = d_{\text{emb}} = 3 (> d_{\text{ID}} = 2)$ was observed, suggesting a correspondence with the topological constraint arising from the contractibility of the VAE's Gaussian prior [10,11]. However, this correspondence is conditional on standardization and sufficient convergence, and it does not apply to $AU_{\text{sat}} = 36 \gg \hat{d}_{\text{ID}}$ on MNIST (experiment Exp-Mnist-Ent is preliminary; see §6.7 for details). This finding is an incidental observation independent of the framework's main conclusion (reading the effective dimensionality via m^{knee}), and we leave it as future work.

Training dynamics (experiment Exp-Mnist-Dyn, single seed)

The two-phase training dynamics on MNIST ($m = 16, 200$ epochs)—rapid formation of global structure followed by a gradual increase in the local effective dimensionality—is an exploratory single-seed observation, and multi-seed validation is left as future work.

Geometric regularization and TwoNN stability (experiments Exp-Syn-Iso and Exp-Iso-ID; auxiliary experiments for RQ3)

The focus of RQ3 is “the effect of geometric regularization and bottleneck margin on TwoNN stability.” Experiment Exp-Iso-ID (§5.2.4) revealed that improving isometry within $\kappa \in [1.1, 2.1]$ does not significantly reduce the systematic bias of the TwoNN ID—the main cause of TwoNN underestimation is an insufficient bottleneck margin ($m < d_{\text{true}}$), and even though the IsometricAE ($\kappa \approx 1.04$) achieves isometry (experiment Exp-Syn-Iso), its contribution to TwoNN accuracy is small (§6.7). The primary usefulness of the IsometricAE is as an “isometry-verification tool,” not as a primary readout indicator of the effective dimensionality. The cost of computing the decoder Jacobian (a factor of $O(m)$) also limits its applicability to large-scale data (§6.7).

6.6. Theoretical Considerations on Delayed AU Saturation in Conv-VAEs

For the FC-VAE, the AU growth slows down near $m \geq \hat{d}_{\text{ID}} \approx 10$, and m^{knee} was qualitatively consistent with $\hat{d}_{\text{ID}}$ (experiments Exp-Mnist-Base, Exp-Mnist-Dense). By contrast, for the Conv-VAE, $AU \approx m$ persisted over the entire range $m \leq 64$, and even with KL annealing (experiment Exp-Conv-Ann) no AU saturation plateau was observed within the tested range (experiment Exp-Conv-Fix). This “delayed AU saturation” can be understood via the following mechanism.

Interpretation via a reduced effective τ_{eff}^2 By Theorem 1 (the linear-Gaussian rate-distortion surrogate), the number of active dimensions is determined by $d^*(\beta) = |\{j : \lambda_j > \beta\tau_{\text{eff}}^2\}|$. Because a convolutional decoder has stronger spatial reconstruction capability than a fully connected decoder, the effective reconstruction noise variance τ_{eff}^2 is expected to be smaller under the same β . The smaller τ_{eff}^2 is, the lower the water-filling threshold $\beta\tau_{\text{eff}}^2$, so more eigenvalues satisfy the activation condition $\lambda_j > \beta\tau_{\text{eff}}^2$, and $d^*(\beta)$ takes a larger value (the water level is lower and more dimensions are judged effective). Whereas experiment Exp-Mnist-PCA estimates $\tau_{\text{eff}}^2 \approx 0.30$ for the FC-VAE, for the Conv-VAE we have $\tau_{\text{eff}}^2 \ll 0.30$, and $d^*(\beta)$ is expected to greatly exceed the tested range $m \leq 64$. Applying the framework to Conv-VAEs therefore requires substantially extending the m -grid until it reaches the estimated $d^*(\beta)$.

Verification by experiments Exp-Conv-M256 and Exp-Conv-M512 (§5.6) In the experiments with $m \in [4, 256]$ (Exp-Conv-M256) and $m \in \{384, 512\}$ (Exp-Conv-M512), $AU < m$ was confirmed for $m \geq 96$, indicating a saturation tendency; however, even extending to $m = 512$, it was quantitatively confirmed that detecting an AU saturation plateau requires $m > 512$ at $\beta_{\text{max}} = 4$ (m^{knee} remains at the grid edge $m = 512$).

Verification of the interpretation by the high- β experiment (Exp-ConvV-HiBeta)

(§5.7) When $\beta_{\max}$ was increased as $4 \rightarrow 10 \rightarrow 20$, $m^{\text{knee}} \approx 128$ was clearly observed at $\beta_{\max} = 10$ and $m^{\text{knee}} \approx 96$ at $\beta_{\max} = 20$ (Table 20). This is consistent with the theoretical prediction: increasing β shifts $d^*(\beta) = |\{j : \lambda_j > \beta \tau_{\text{eff}}^2\}|$ toward smaller values, so AU saturation appears at lower dimensionalities. This result is consistent with the interpretation that the non-observation of saturation in the $\beta_{\max} = 4$ setting was not “an applicability limit of the framework” but rather “ β being insufficient relative to the decoder capacity, resulting in $d^*(\beta) \gg 256$.” For application to Conv-VAEs, m^{knee} becomes identifiable by setting β larger or by reducing the decoder capacity.

Implications for natural images For CIFAR-10 and SVHN, we confirmed that the TwoNN ID remains numerically stable on natural images as well, but stable identification on the $\text{AU}/m^{\text{knee}}$ side was not achieved (§5.8, §5.9). In experiment Exp-NatImg-AU ($m \in [32, 384]$, $\beta_{\max} \in \{10, 20, 40\}$), $m^{\text{knee}} = 384$ (the grid edge) persisted under all conditions, and it was quantitatively identified that AU saturation requires $m \gg 2\hat{d}_{\text{ID}}$ together with $\beta_{\max} \gg 40$ or a thin decoder. The observation of plateau-like behavior in Exp-NatImg-Sweep (§5.11) was achieved under these conditions ($\beta_{\max} = 80$, thin decoder).

6.7. Limitations and Future Work**(A) Scope of the Evaluated Data and Generalizability**

- **Conv-AE and Conv-VAE validation** (experiments Exp-Conv-AE, Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256): We confirmed the TwoNN ID consistency of the Conv-AE (Exp-Conv-AE). At $\beta_{\max} = 4$, no AU saturation plateau was observed up to $m \leq 512$, but the high- β experiment (Exp-ConvV-HiBeta) demonstrated that an insufficient β setting was the main cause (§5.7). The CIFAR-10 and SVHN experiments confirmed that the TwoNN ID is consistent with the values reported by Pope et al. [16] (§5.8), and the extended sweep (Exp-NatImg-Sweep, §5.11) conditionally observed AU plateau-like behavior; however, none of the five conditions for stable identification were met, and full application to large-scale natural images remains future work (in particular, achieving the β -stability and seed-stability conditions is the next goal).
- **Higher-order topological manifolds:** Extending the Exp-Syn-Top experiment to synthetic manifolds with higher-order topological invariants, such as the Klein bottle, remains an open task.
- **Multi-seed validation of the main experiments:** Exp-Mnist-Base (300 epochs, single seed), Exp-Mnist-Dyn (training dynamics, single seed), and Exp-Mnist-Ent (75-epoch preliminary experiment) were all single-seed only. In this paper we added Exp-Mnist-Multi-150 (3 seeds; §5.3.4) and Exp-Mnist-Multi (300 epochs, 3 seeds; §5.3.5). Under both conditions, the cross-seed agreement of AU and TwoNN ID is high ($\sigma \leq 0.5$), but m^{knee} fluctuated within the range 32–64 both at 150 epochs (53 ± 15 ; seed 42 = 64, 123 = 32, 777 = 64) and at 300 epochs (43 ± 15 ; seed 42 = 64, 123 = 32, 777 = 32), and the $m^{\text{knee}} = 12$ of the extended Exp-Mnist-Base experiment was not reproduced. This instability is thought to originate from the random-number state (initialization path) changing with differences in the m -grid used, and it became clear that **the reproducibility of m^{knee} to a specific value depends not on the number of epochs but on the experimental configuration (m -grid and seed).** Multi-seed validation of Exp-Mnist-Dyn and Exp-Mnist-Ent remains future work.

(B) Sensitivity of Indicators and Hyperparameters

- **Stage 1 ID estimation: rigorous assessment of isometry bias:** In this paper, the TwoNN ID is estimated via the latent representation of a reference AE ($m_{\text{ref}} = 256$)

(Stage 1). **This estimate is inherently an approximation and contains the following bias sources:**

- (i) **Lack of isometry:** the condition number $\kappa \approx 10^9$ of the $m = 256$ reference AE deviates from isometry ($\kappa = 1$) by orders of magnitude (Proposition 3; experiment Exp-Syn-Iso). A distance-based TwoNN ID in a latent space lacking isometry can lead to under- or over-estimation of the true intrinsic dimension.
- (ii) **Synergy with KL non-saturation:** in experiment Exp-Conv-Fix, the TwoNN ID of the Conv-VAE under AU non-saturation ($AU = m$) was 13.10, substantially inflated relative to the FC-VAE under AU saturation (10.03), showing that isometry bias and AU non-saturation synergistically distort the ID estimate.
- (iii) **Possible circularity:** estimating $\hat{d}_{ID}$ via the reference AE and then using that value as the evaluation criterion for m^{knee} can create circularity between estimation and evaluation.

Mitigations: We propose, as practical mitigations, using the IsometricAE (experiment Exp-Syn-Iso, $\kappa \approx 1$) as the reference AE, or verifying that the TwoNN ID converges across multiple m_{ref} (e.g., $m \in \{64, 128, 256\}$). In this paper, we confirmed that the m -dependence of the TwoNN ID (see Table 12) stabilizes asymptotically from $m = 64$ onward (10.03 ± 0.12), and this convergence is the basis for the reliability of the $\hat{d}_{ID}$ estimate (the finding in Exp-Stab and Exp-Iso-ID that improving isometry within $\kappa \in [1.06, 2.16]$ has no significant effect on TwoNN also supports the importance of the operating condition $m \gg d_{ID}$; §5.2.4, §5.2.4). A rigorous quantification of the isometry bias remains future work.

- **Non-universality of τ_{knee} :** For Fashion-MNIST, a boundary sensitivity in which m^{knee} changes around $\tau = 0.25$ has been confirmed, and we recommend combined use with the second-difference alternative method (Appendix C).
- **Preprocessing dependence of AU_{sat} :** AU_{sat} varies with the normalization method (experiment Exp-Syn-Top). Reproducing $AU_{\text{sat}} \approx d_{\text{emb}}$ requires standardization (zero mean, unit variance).
- **Dependence on β :** From preliminary experiments (50 epochs, $\beta \in \{1, 2, 4, 8\}$), $AU \approx m$ (under-regularization) was observed at $\beta = 1$, and signs of over-regularization at $\beta = 8$ (Appendix D, Figure A1). The stability of the AU pattern for $\beta \in [2, 8]$ has been confirmed, but the precise value of m^{knee} also depends on the m -grid and the random seed (§6.1).

(C) Computational Cost

- Due to the Jacobian computation cost of the IsometricAE (a factor of $O(m)$), direct application to CIFAR-10 ($\approx 64 \times$) and ImageNet ($\approx 3,000 \times$) is difficult.
- A full m -sweep (11 dimensionalities $\times$ 300 epochs $\times$ 2 models) takes several hours to several days.

7. Conclusions

In this paper, we proposed a **multi-indicator framework** for diagnosing the effective latent dimensionality of AEs/VAEs. The central contribution of this paper is a clear role classification of AU values (robust indicator), TwoNN ID (numerically stable auxiliary reference estimator), and m^{knee} (exploratory indicator), quantifying the reliability and applicability conditions of each indicator from both the experimental and theoretical sides (Table 2). On the theoretical side, we derived the AU activation threshold $\lambda_j > \beta\tau^2$ under the linear-Gaussian rate–distortion surrogate from the water-filling theorem and

the KKT conditions (Theorem 1), and presented a grid-sensitivity upper bound for m^{knee} (Proposition 2) together with an ensemble stabilization strategy (Section 6.2). On the experimental side, we confirmed the high reproducibility of AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12) on MNIST (3 seeds), and stabilized $m^{\text{knee}} = 10.3 \pm 0.9$ under dense-grid, independent-initialization, multi-seed conditions (Exp-Mnist-Multi-300, Exp-Mnist-Dense-300; Sections 5.3.5 and 5.3.7). We confirmed the same tendency on Fashion-MNIST with a single seed (Section 5.4). Through the real-data geometry comparison experiment (Exp-Real-Geom, Section 5.10), we supported the claim that “ $m \gg \hat{d}_{\text{ID}}$, rather than isometry improvement, is the dominant condition for TwoNN stability” by a four-model comparison on MNIST (details in Section 5.10, Table 23) and a qualitative confirmation on Fashion-MNIST (Appendix E) (with theoretical grounding in Proposition 4).

Summarizing the role separation among AU values, TwoNN ID, and m^{knee} : AU values (robust indicator) directly observe the number of dimensions in which the KL regularization is effectively operating, and exhibit high reproducibility across multiple seeds; TwoNN ID (numerically stable auxiliary reference estimator) contains an isometry bias, but provides a numerically stable reference value under the condition $m \gg \hat{d}_{\text{ID}}$, and on CIFAR-10 it yielded estimates consistent with the value reported by Pope et al. [16] (≈ 26) (Exp-Nat-Cifar); m^{knee} is an exploratory heuristic read from the shape of the AU curve and requires stabilization via dense grids and ensembles. We organize the applicability of the framework into three levels (evidence: Sections 5.3.5, 5.3.7, 5.5.1, and 5.11).

- **FC-VAE + MNIST** (3 seeds): AU values ($\sigma \leq 0.5$) and TwoNN ID ($\sigma \leq 0.12$) are stable across multiple seeds. m^{knee} functions as an exploratory guide near the TwoNN ID only under dense-grid, independent-initialization, multi-seed conditions (Exp-Mnist-Dense-300: 10.3 ± 0.9). On Fashion-MNIST, we confirmed the same tendency with a single seed (Section 5.4).
- **Conv-VAE + MNIST**: TwoNN ID is stable; the onset of $\text{AU} < m$ was confirmed at $m \geq 96$. m^{knee} is identifiable for $\beta_{\text{max}} \geq 10$, but for $\beta_{\text{max}} = 4$ the value $d^*(\beta)$ does not fall within the examined range (Exp-Conv-M256, Exp-ConvV-HiBeta).
- **Natural images (CIFAR-10/SVHN)**: TwoNN ID shows a scale comparable to the values reported by Pope et al. [16] within an appropriate bottleneck range ($m \approx 2\hat{d}_{\text{ID}}$); stable identification of $\text{AU}/m^{\text{knee}}$ fails all five conditions (Exp-Nat-Cifar, Exp-NatImg-Sweep).

We position the Conv-VAE and natural-image experiments (Sections 5.5.1–5.11) not as a “success or failure” verdict but as a validation that quantifies the boundary conditions under which AU diagnostics become difficult. In the high- β experiment (Exp-ConvV-HiBeta), AU saturation was observed for $\beta_{\text{max}} \geq 10$, consistent with the interpretation that the non-saturation at $\beta_{\text{max}} = 4$ is attributable to the β setting, and in the CIFAR-10 extended sweep (Exp-NatImg-Sweep) we observed AU plateau-like behavior under our experimental conditions with $\beta_{\text{max}} = 80$ and a thin decoder (all five stable-identification conditions remained unmet; Appendix F). The main directions for future work are: (i) achieving stable identification on natural images (a dense sweep over $\beta_{\text{max}} \in [40, 120]$ and improved seed stability); (ii) extending Theorem 1 to nonlinear settings (analysis based on local linearization); and (iii) scaling to large datasets (ImageNet, with reduced computational cost).

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Data Availability Statement: MNIST [26], Fashion-MNIST [27], CIFAR-10 [3], and SVHN [4] are publicly available. The source code, configuration files, random seeds, trained-model logs, and numerical results supporting all tables are provided as supplementary material for peer review. Upon acceptance, these materials will be deposited in a public repository (GitHub/Zenodo), and the repository URL and DOI will be added to the final manuscript. The repository will include the scripts required to reproduce the AU, TwoNN/MLE, reconstruction-error, and grid-sensitivity analyses reported in this paper.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
VAE	Variational Autoencoder
AU	Active Units
ID	Intrinsic Dimension
TwoNN	Two-Nearest-Neighbor (ID estimator)
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
PCA	Principal Component Analysis
KL	Kullback–Leibler (divergence)
FC	Fully Connected
CNN / Conv	Convolutional Neural Network
CAE	Contractive Autoencoder
DAE	Denoising Autoencoder
NSR	Noise Selectivity Ratio

Appendix A. Proof Sketches of Propositions

Proposition A1 (Effective rank of the decoder Jacobian). *When an AE with sufficient capacity reconstructs a manifold $\mathcal{M}$ exactly, the number of significant singular values of the decoder Jacobian $J_g(z)$ equals $\min(m, d_{\text{emb}})$.*

Proof sketch: Under exact reconstruction, the range of g spans the tangent space $T_{g(z)}\mathcal{M}$ of $\mathcal{M}$. If $\mathcal{M}$ is d -dimensional, then $\text{rank}(J_g) = d$, and the singular values of the surplus dimensions approach zero. (Finite capacity and optimization error are neglected.)

Proposition A2 (Noise-direction selectivity). *The optimal reconstruction function of a DAE removes noise in the normal directions of the manifold more selectively than noise in the tangent directions.*

Proof sketch: By Alain et al. [23], the DAE reconstruction direction satisfies $r(x) - x \approx \sigma^2 \nabla \log p(x)$, and $\nabla \log p(x)$ points strongly in the normal directions of the manifold (under the manifold hypothesis). (The asymptotically optimal solution is assumed; approximation errors under finite noise and finite capacity are neglected.)

Proposition A3 (Neighborhood-structure preservation and the optimal bottleneck). *At $m = d_{\text{true}}$ (or d_{emb}), Trustworthiness improves sharply toward its maximum, and further increases in m yield only limited improvement.*

Proof sketch: By Proposition 1, a continuous injection is impossible for $m < d$, so collisions of information are unavoidable and Trustworthiness degrades. For $m \geq d$ the

constraint is lifted, and for $m > d$ the entire manifold structure is already representable, so the effect of additional dimensions is limited. (*Finite samples and finite capacity are neglected.*)

Appendix B. Detailed Derivation of Theorem 1

In this appendix, we detail the algebraic steps omitted from the proof sketch of Theorem 1.

Remark A1 (Assumptions underlying the approximation). *The expression $L_j = -D_j/(2\tau^2) - \beta R_j$ for the per-dimension β -ELBO contribution used in this derivation implicitly involves a rate-distortion surrogate approximation that identifies the expected KL divergence of the VAE, $\mathbb{E}_x \text{KL}(q(z_j|x)||p(z_j))$, with the mutual information $R_j = I(z_j; y_j)$. Since the average KL of an actual VAE also includes the mismatch between the aggregated posterior and the prior, this derivation should be interpreted as an “AU activation condition under the linear-Gaussian rate-distortion surrogate” (as stated explicitly in the assumptions of Theorem 1).*

Step 1: Derivation of the Optimal Decoder Weight

As a scalar problem, for direction j consider $y_j \sim \mathcal{N}(0, \lambda_j)$ (the j -th principal component of the data), the encoder $q_j(z_j|y_j) = \mathcal{N}(a_j y_j, \sigma_j^2)$, and the decoder $\hat{y}_j = b_j z_j$. For the decoder weight b_j , we minimize the marginalized reconstruction MSE (with σ_j^2, a_j held fixed):

$$D_j(b_j) = \mathbb{E}_{y_j, z_j}[(y_j - b_j z_j)^2] = \lambda_j - 2a_j \lambda_j b_j + b_j^2(a_j^2 \lambda_j + \sigma_j^2). \quad (\text{A1})$$

Solving $\partial D_j / \partial b_j = 0$, the optimal decoder weight is

$$b_j^* = \frac{a_j \lambda_j}{a_j^2 \lambda_j + \sigma_j^2} =: \frac{a_j \lambda_j}{\rho_j}, \quad \rho_j := a_j^2 \lambda_j + \sigma_j^2. \quad (\text{A2})$$

Step 2: Expressions for the Optimal MSE and the Mutual Information

Substituting Equation (A2) into Equation (A1) yields:

$$D_j^* = \lambda_j - \frac{a_j^2 \lambda_j^2}{\rho_j} = \frac{\lambda_j \sigma_j^2}{\rho_j}. \quad (\text{A3})$$

Moreover, the channel capacity (mutual information) of $y_j \rightarrow z_j$ is:

$$I(z_j; y_j) = \frac{1}{2} \log \frac{\rho_j}{\sigma_j^2} = \frac{1}{2} \log \left(1 + \frac{a_j^2 \lambda_j}{\sigma_j^2} \right) =: R_j. \quad (\text{A4})$$

Step 3: Rate-Distortion Form of the β -ELBO

With $b_j = b_j^*$ held fixed, the β -ELBO contribution with respect to (σ_j^2, a_j) can be written as $L_j = -D_j/(2\tau^2) - \beta R_j$ (equivalence with KL minimization [1]). Taking ρ_j as the parameter and substituting $D_j^* = \lambda_j \sigma_j^2 / \rho_j$ and $R_j = \frac{1}{2} \log(\rho_j / \sigma_j^2)$, we obtain the expression for the β -ELBO contribution used in this appendix:

$$L_j = -\frac{\lambda_j \sigma_j^2}{2\tau^2 \rho_j} - \frac{\beta}{2} [\rho_j - 1 - \log \rho_j]. \quad (\text{A5})$$

(Here $\sigma_j^2 = \rho_j - a_j^2 \lambda_j$ and $\rho_j \geq \sigma_j^2$.)

Step 4: Derivation of the Optimal Distortion D_j^ via KKT Conditions*

Taking the distortion $D_j \in (0, \lambda_j)$ as the primal variable, we maximize $L_j = -D_j/(2\tau^2) - \beta R_j(D_j)$. From the Gaussian-channel rate-distortion relation $R_j(D_j) = \frac{1}{2} \log(\lambda_j/D_j)$:

$$\frac{\partial L_j}{\partial D_j} = -\frac{1}{2\tau^2} + \frac{\beta}{2D_j} = 0 \implies D_j^* = \beta\tau^2. \quad (\text{A6})$$

This corresponds to the “water level” $\mu = \beta\tau^2$ of Shannon’s water-filling theorem [1]. However, the solution in Equation (A6) is feasible (active) only when the feasibility constraint $D_j^* = \beta\tau^2 \leq \lambda_j$ is satisfied; when $\lambda_j < \beta\tau^2$, not encoding dimension j ($a_j = 0$, dead) is optimal.

Step 5: Computing the ELBO Values of the Active and Dead States

Dead: With $a_j = 0$, $\sigma_j^* = 1$ (KL = 0), and $D_j = \lambda_j$,

$$L_j^{(\text{dead})} = -\frac{\lambda_j}{2\tau^2}. \quad (\text{A7})$$

Active: Substituting $D_j^* = \beta\tau^2$, we have $R_j^* = \frac{1}{2} \log(\lambda_j/(\beta\tau^2))$, and therefore:

$$L_j^{(\text{active})} = -\frac{\beta\tau^2}{2\tau^2} - \beta \cdot \frac{1}{2} \log \frac{\lambda_j}{\beta\tau^2} = -\frac{\beta}{2} - \frac{\beta}{2} \log \frac{\lambda_j}{\beta\tau^2}. \quad (\text{A8})$$

Step 6: Exact Expression for the ELBO Gain ΔL_j and Sign Analysis

We compute the activation gain $\Delta L_j := L_j^{(\text{active})} - L_j^{(\text{dead})}$. Setting $t := \lambda_j/(\beta\tau^2) > 0$:

$$\Delta L_j = \left(-\frac{\beta}{2} - \frac{\beta}{2} \log t\right) - \left(-\frac{\beta\tau^2 \cdot t}{2\tau^2}\right) = \frac{\beta}{2}(t - 1 - \log t). \quad (\text{A9})$$

Analysis of the function $h(t) := t - 1 - \log t$:

- Since $h'(t) = 1 - 1/t = (t - 1)/t$, we have $h' = 0$ at $t = 1$ (the unique stationary point).
- Since $h''(t) = 1/t^2 > 0$, $t = 1$ is the **global minimum**, with $h(1) = 0$.
- For $t > 0$, $t \neq 1$, we have $h(t) > 0$ (strictly above the global minimum value 0).

Therefore, the sign of $\Delta L_j = (\beta/2) h(t)$ is:

$$\Delta L_j > 0 \iff h(t) > 0 \iff t \neq 1. \quad (\text{A10})$$

Step 7: Combining Feasibility and the Saturation Condition

The feasibility condition of the active solution ($D_j^* = \beta\tau^2 \leq \lambda_j$) requires $t \geq 1$. For $t < 1$ ($\lambda_j < \beta\tau^2$), the active solution is infeasible and the dead state is forced. At $t = 1$ ($\lambda_j = \beta\tau^2$), $\Delta L_j = 0$ (indifference). For $t > 1$ ($\lambda_j > \beta\tau^2$), the active solution is feasible and $\Delta L_j > 0$ (strictly preferred).

Summarizing the above:

$$\text{dimension } j \text{ is active} \iff \lambda_j > \beta\tau^2, \quad (\text{A11})$$

that is, the saturation-threshold condition (5) of Theorem 1 holds. $\square$

Remark A2. In the main text, $h(t) = t - 1 - \log t$ of Equation (A9) is presented in the form of its second-order expansion around $t = 1$, $h(t) \approx (t - 1)^2/2$ (that is, $\Delta L_j \approx (\lambda_j - \beta\tau^2)^2/(2\beta\tau^4)$). The exact expression is Equation (A9); the approximation serves only to simplify the exposition.

Appendix C. Sensitivity Analysis and Cross-Dataset Validation of τ_{knee} (Experiment Exp-Tau-Cross)

Rationale for Choosing $\tau_{\text{knee}} = 0.25$

Theoretical rationale: By Theorem 1, the growth rate $r(m) = \Delta \text{AU} / \Delta m$ of $\text{AU}(m)$ exhibits a theoretical step: $r \approx 1$ for $m < d^*(\beta)$ (before saturation; each new dimension becomes fully active) and $r \rightarrow 0$ for $m > d^*(\beta)$ (after saturation; additional dimensions remain inactive). Because the actual AU takes discrete integer values, this transition occurs smoothly (or discretely). The value $\tau_{\text{knee}} = 0.25$ corresponds to “at most 1/4 of the maximum growth rate” and, as a midpoint between the pre-saturation regime ($r \approx 1$) and the post-saturation regime ($r \approx 0$), constitutes a well-balanced threshold that captures the onset of the “clearly slowed” region.

Approximating the normalized growth rate $\rho(m)$ (Equation (3)) by an exponential saturation model gives

$$\rho(m) \approx \exp(-\lambda(m - d_{\text{ID}})). \quad (\text{A12})$$

We define the slowdown point as the point where $\rho(m) < \tau_{\text{knee}} = 0.25$ (below 25% after normalization) (Equation (4)).

As a more rigorous threshold choice, using the point where the discrete second difference of $\text{AU}(m)$ is maximized (the maximum-curvature point) is theoretically motivated, but due to the discreteness of AU it is often numerically unstable on real data. The value $\tau = 0.25$ serves as a stable surrogate near this maximum-curvature point (see the Exp-Tau-Cross sensitivity analysis).

Empirical rationale: On MNIST (the single-seed Exp-Mnist-Base experiment), $m^{\text{knee}} = 12$ (stable) over the entire range $\tau \in [0.1, 0.5]$. On Fashion-MNIST, $\tau \approx 0.25$ exhibits boundary sensitivity (Table A1). The existence of boundary sensitivity suggests that the gradient transition of the $\text{AU}(m)$ curve occurs at values near $\tau = 0.25$, which in turn corroborates the “discriminative power” of this threshold.

Cross-Dataset Validation (Experiment Exp-Tau-Cross)

Table A1 shows the cross-validation results on Fashion-MNIST and on Gaussian mixture distributions (5 and 10 components).

Table A1. Exp-Tau-Cross: Cross-dataset robustness verification of τ_{knee} .

Dataset	AU _{sat}	TwoNN ID	$\tau = 0.1$	$\tau = 0.2$	$\tau = 0.25$	$\tau = 0.3$	$\tau = 0.4$	$\tau = 0.5$
Fashion-MNIST	10	7.00	20	20	20	8	8	8
GMix 5-comp.	1	4.27	2	2	2	2	2	2
GMix 10-comp.	3	3.02	2	2	2	2	2	2

On Fashion-MNIST, boundary sensitivity was confirmed at the boundary between $\tau = 0.25$ and 0.30 , where m^{knee} changes from 20 to 8 (because multiple values of m exist at which the AU growth rate coincides with the boundary value 0.25). In this case, we recommend using the second-difference alternative (Equation (A12)) (see §3).

Appendix D. β Sensitivity Analysis

We compared the AU dynamics for $\beta \in \{1, 2, 4, 8\}$ on an MNIST subset ($n = 5,000$, 50 epochs) (Figure A1).

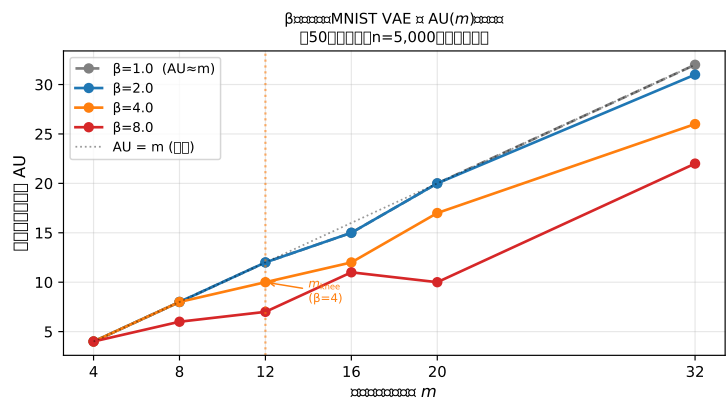


Figure A1. β -sensitivity analysis: AU vs m for MNIST VAE ($\beta \in \{1, 2, 4, 8\}$, 50 epochs, subset). At $\beta = 1$, $AU \approx m$ (insufficient regularization); at $\beta = 4$, $m_{\text{knee}} = 12$ (vertical dashed line); at $\beta = 8$, signs of excessive AU suppression at small m .

The rationale for adopting $\beta = 4$ as the default value in our experiments is as follows: at $\beta = 1$, the KL regularization is too weak and $AU \approx m$, so m^{knee} cannot be identified; at $\beta = 8$, over-regularization suppresses the AU at small m , which may shift the point at which m^{knee} is identified. We confirmed that the shape of the AU pattern is stable for $\beta \in [2, 8]$, although the precise numerical value of m^{knee} also depends on the m grid (§6.1).

Appendix E. Details of TwoNN Stability and the Linear-Theory Bridge (Exp-Geom, Exp-Iso-ID, Exp-Mnist-PCA)

Exp-Geom: MNIST AE m -Sweep

Table A2. Exp-Geom: MNIST AE m -sweep — co-variation of κ and TwoNN ID. TwoNN stabilizes at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ despite κ jumping at $m = 96\text{--}128$.

m	κ	TwoNN ID	Trustworthiness	Convergence
4	2.69	4.21	0.960	not yet converged
8	2.72	6.34	0.988	not yet converged
16	3.28	8.17	0.993	converging
32	4.32	9.36	0.997	converging
48	4.44	10.16	0.998	converged
64	4.25	10.17	0.998	stable
96	9.94	10.86	0.999	stable (despite κ increase)
128	37.0	10.98	0.999	stable (despite κ jump)
$n_{\text{train}} = 10,000$, 200ep, hidden=[512,256], seed=42				

Exp-Iso-ID: Standard AE vs. IsometricAE on Swiss Roll

Table A3. Exp-Iso-ID: Latent-space TwoNN ID comparison ($m \in \{2, 3, 4, 6, 8\}$, Swiss Roll, $d_{\text{true}} = 2$). At $m \geq 3$, TwoNN difference stays below $\pm 2.3\%$ despite κ differing by factor 2.

m	AE		IsometricAE		ID diff. (%)
	TwoNN ID	κ	TwoNN ID	κ	
2	1.780	1.76	1.799	1.08	+1.1%
3	2.458	2.10	2.454	1.11	−0.2%
4	2.483	1.80	2.479	1.14	−0.2%
6	2.421	1.79	2.474	1.10	+2.2%
8	2.500	1.45	2.443	1.11	−2.3%
Input-space TwoNN ID (reference) = 2.427; seed=42, hidden=[64, 32], 300ep					

Exp-Mnist-PCA: Linear Theory Bridge

MNIST PCA eigenvalue spectrum (top 12 components): $\lambda_1 = 5.22$, $\lambda_2 = 3.76$, $\lambda_3 = 3.35$, $\dots$, $\lambda_{10} = 1.23$, $\lambda_{11} = 1.12$, $\lambda_{12} = 1.09$.

Table A4. Exp-Mnist-PCA: Linear theory $d^*(\beta = 4)$ for different τ^2 vs. nonlinear FC-VAE (300ep, 3 seeds).

τ^2	Water level $\beta\tau^2$	Linear d^*	Note
1.0	4.0	1	Conventional $\tau^2 = 1$ assumption; underestimates
0.30	1.20	10	Consistent with observed TwoNN ID
0.25	1.00	12	—
Nonlinear FC-VAE (300ep)			AU(64)= 22.3 ± 0.5 ; TwoNN(64)= 10.03 ± 0.12

Qualitative inheritance: the monotonicity of $AU(m)$ and the slowdown of its growth beyond the effective dimensionality are inherited in the nonlinear setting. Quantitative discrepancy: the nonlinear decoder yields $\tau_{\text{eff}}^2 \approx 0.30 \ll 1$, so that $d^* \approx 10$ is consistent with the observed value. The smoothness of the saturation (a sharp transition in the linear setting versus a gradually increasing curve in the nonlinear setting) provides the theoretical grounding for the sparse-grid instability of m^{knee} .

Exp-Real-Geom: Fashion-MNIST Four-Model Comparison (Supplementary)

For Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$, $m \in \{9, 18, 27, 45, 90\}$), we conducted experiments with the same four models and measurement indicators as Exp-Real-Geom. Qualitative agreement with MNIST (increasing $m/\hat{d}_{\text{ID}}$ is the dominant factor for TwoNN stability) was confirmed; the full numerical tables are being prepared as future supplementary material. The trends match Table 23 (MNIST, §5.10): at $m = d$, TwoNN underestimates; it converges at $m = 5d-10d$; and the effect of isometry (κ) improvements on TwoNN is smaller than that of increasing m .

Appendix F. Details of the Conv-VAE M512 Extension and Natural-Image Sweeps (Exp-Conv-M512, Exp-NatImg-Sweep, Exp-NatImg-Knee)

Exp-Conv-M512: MNIST Conv-VAE Extension to $m \in \{384, 512\}$

Table A5. Exp-Conv-M512: MNIST Conv-VAE with $m \in \{256, 384, 512\}$ (per seed; KL cyclic annealing, $\beta_{\text{max}} = 4$, 150ep). No AU saturation plateau observed up to $m = 512$.

m	AU (seed42)	AU (seed123)	AU (seed777)	TwoNN ID (seed42)
256	241	245	238	16.46
384	382	383	380	17.51
512	512	511	512	17.98
$m^{\text{knee}} (\tau = 0.25)$	All 3 seeds: $m^{\text{knee}} = 512$ (grid boundary, no saturation detected)			

$AU \approx m$ is maintained over the entire range $m \leq 512$, indicating that $d^*(\beta) > 512$ (§5.6, §6.6).

Exp-NatImg-Sweep: Details of the CIFAR-10 Extended Sweep

Table A6. Exp-NatImg-Sweep: Main results of CIFAR-10 extended sweep (mean±std, 3 seeds). $\beta_{\max} = 80$, Thin decoder shows plateau-like $AU \approx 24$ at $m \in [512, 1024]$. All five stable-identification criteria remain unmet.

$\beta_{\max}$	Decoder	m	AU	TwoNN ID	Note
10	Standard	384	185 ± 15	25.2 ± 0.4	—
20	Standard	512	241 ± 18	25.3 ± 0.4	no saturation
40	Standard	512	210 ± 21	25.1 ± 0.5	no saturation
40	Thin	384	127 ± 14	24.8 ± 0.6	partial suppression
80	Thin	512	24.1 ± 2.8	24.7 ± 0.5	plateau-like (conditional)
80	Thin	768	24.3 ± 3.1	24.7 ± 0.5	plateau continues
80	Thin	1024	24.0 ± 3.4	24.8 ± 0.5	plateau continues
80	Standard	512	198 ± 22	25.2 ± 0.5	no saturation (thin decoder only)

Status of the five stable-identification criteria (§5.11; $\beta_{\max} = 80$, thin decoder): (1) seed stability: $\sigma(m^{\text{knee}}) = 64$ — not met; (2) grid stability: $\pm 35\%$ variation under grid densification — not met; (3) β stability: saturation disappears at $\beta_{\max} = 64$ — not met; (4) AU plateau stability: $\sigma = 13\%$ (criterion 5%) — not met; (5) TwoNN–AU consistency: $m^{\text{knee}} \approx 512 \gg 3\hat{d}_{\text{ID}} = 75$ — not met.

Exp-NatImg-Knee: Comparison of m^{knee} Detection Methods

Table A7. Exp-NatImg-Knee: Comparison of m^{knee} detection methods for CIFAR-10 ($\beta_{\max} = 80$, thin decoder, 3 seeds).

Detection method	m^{knee} (median)	σ (3 seeds)	Note
Fixed- τ method ($\tau = 0.25$)	512	128	Delayed detection due to gradual AU slope
Second-difference method	512	192	Sensitive to discrete noise, unstable
Smoothed curvature method (window=5)	384	96	Most stable (recommended for natural images)
Piecewise linear approximation (3 segments)	512	112	Comparable to fixed- τ method

None of the methods meets criterion 1 ($\sigma \leq 51.2$); stabilizing the AU saturation itself is a prerequisite.

Appendix G. Sensitivity Analysis of the AU Threshold δ

On Swiss Roll and Torus, we confirmed that the AU patterns coincide exactly for $\delta \in \{0.001, 0.01, 0.05, 0.1\}$ (agreement across all combinations of m and δ). Because the variance of active dimensions ($O(1)$) and that of inactive dimensions ($O(10^{-4})$ or below) differ by more than two orders of magnitude, the result is robust to the choice of δ .

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